

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\403-E-003.pvdb

Date of Analysis : Jul 4, 2026 2:18pm

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 490.000 mm. ID
 Vessel Design Length, Tangent to Tangent 6811.30 mm.
 Specified Datum Line Distance 50.00 mm.
 Shell Side Design Temperature 132 °C
 Channel Side Design Temperature 77 °C
 Shell Side Design Pressure 17.237 bars
 Channel Side Design Pressure 5.516 bars
 Shop Shell Side Test Pressure 36.361 bars
 Shop Channel Side Test Pressure 36.361 bars
 Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
 Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-266 4	...	...	K03017	
Nozzle	SA-106 B	...	...	K03006	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-179	...	...	K01200	
Tubesheet	SA-266 4	...	...	K03017	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Fr. END COVER	5.564	0.00	50.000	3.1750	No	No
CH.SHELL	5.564	0.00	48.000	3.1750	N/A	No
CH FLG	5.564	0.00	29.000	3.1750	N/A	No
Shell FLG	17.285	0.00	29.000	3.1750	N/A	No
Main shell	17.285	0.00	48.000	3.1750	N/A	No
Rear End Cover	17.285	0.00	50.000	3.1750	No	No

Liquid Level: 490.00 mm. Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Ellipse	0.0	50.0	14.0	12.0	5.7	...	1.00	1.00
Cylinder	400.0	400.0	12.0	12.0	5.7	...	1.00	1.00
Body Flg	482.5	78.0	60.0	60.0	57.8	57.8	1.00	1.00
Body Flg	614.5	80.0	60.0	60.0	58.0	58.0	1.00	1.00
Cylinder	6710.5	6096.0	12.0	12.0	6.3	...	1.00	1.00



Ellipse	6761.3	50.8	14.0	12.0	6.2	...	1.00	1.00
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Local Stress Analysis Results:

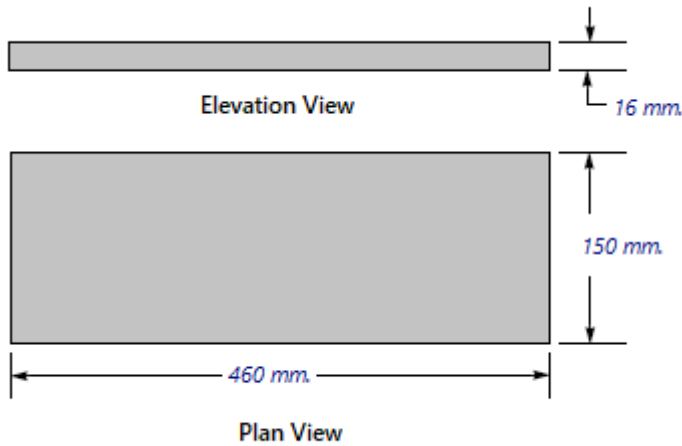
Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
TS IN NOZ T1	WRC-297	0.616	Nozzle	Passed
TS OUT NOZ T2	WRC-297	0.616	Nozzle	Passed
SH IN NOZ S1	WRC-297	0.834	Shell	Passed
SH OUT NOZ S2	WRC-297	0.984	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	150.000 mm.
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	500.000 mm.
Wear Pad Width	225.000 mm.
Wear Pad Thickness	12.000 mm.
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	1220.000 mm.
Baseplate Length	460.000 mm.
Baseplate Thickness	16.000 mm.
Baseplate Width	150.000 mm.
Number of Ribs (including outside ribs)	2
Rib Thickness	12.000 mm.
Web Thickness	12.000 mm.
Height of Center Web	306.000 mm.
Number of Bolts in Baseplate	2

Baseplate Sketch



Maximum Tensile Bolt Load 105. Kg

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	4186.19 Kg
Maximum Transverse Saddle Shear Load	508.02 Kg
Maximum Longitudinal Saddle Shear Load	354.13 Kg

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	3132.30 Kg
Maximum Transverse Saddle Shear Load	167.65 Kg
Maximum Longitudinal Saddle Shear Load	21.95 Kg

Weights:

Fabricated - Bare w/o removable internals	2933.3 kg.
Shop Test - Fabricated + water (full)	4097.2 kg.



Shipping	- Fab. + rem. intls.+ shipping app.	3010.1	kg.
Erected	- Fab. + rem. intls.+ insul. (etc)	3010.1	kg.
Empty	- Fab. + intls. + details + wghts.	3010.1	kg.
Operating	- Empty + operating liquid (no ca)	4172.4	kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
TS IN NOZ T1	48	...	...	OK	5.67	OK	Passed
TS OUT NOZ T2	48	...	...	OK	5.67	OK	Passed
SH IN NOZ S1	44	...	...	OK	6.30	OK	Passed
SH OUT NOZ S2	46	...	...	OK	6.31	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 44 Nozzle : SH IN NOZ S1 [Shellside]
 Minimum MAWP Nozzles : 48 Nozzle : TS IN NOZ T1 [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm.	Layout Angle deg	Dia. Limit mm.
20	TS IN NOZ T1	250.000	-90.000	305.359
20	TS OUT NOZ T2	250.000	90.000	305.359
50	SH IN NOZ S1	6510.500	90.000	305.359
50	SH OUT NOZ S2	914.500	-90.000	400.050

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
TS IN NOZ T1	6.000 in	80	WNF	6.625	10.973	268.27	12.00	176.90	300
TS OUT NOZ T2	6.000 in	80	WNF	6.625	10.973	268.27	12.00	176.90	300
SH IN NOZ S1	6.000 in	80	WNF	6.625	10.973	...	...	187.17	300
SH OUT NOZ S2	8.000 in	80	WNF	8.625	12.700	...	...	212.39	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
TS IN NOZ T	SA-106 B	...	10.000	10.000	12.000	...
TS OUT NOZ	SA-106 B	...	10.000	10.000	12.000	...
SH IN NOZ S	SA-106 B	10.000	10.000	...	...	...
SH OUT NOZ	SA-106 B	10.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
TS IN NOZ T1	200.000	-90.0	150.00	0.00	CH.SHELL
TS OUT NOZ T2	200.000	90.0	150.00	0.00	CH.SHELL
SH IN NOZ S1	6460.500	90.0	150.00	0.00	Main shell
SH OUT NOZ S2	864.500	-90.0	150.00	0.00	Main shell

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	ELLIPTICAL HEAD: 2.0 X 1, 14.0mm. THK X 490.0mm. ID X 50.0mm.	SA-516 70
1	CYLINDER: 12.0mm. THK X 490.0mm. ID X 400.0mm.	SA-516 70
2	WELD NECK FLANGE: 60.0mm. THK X 622.0mm. OD	SA-266 4
1	CYLINDER: 12.0mm. THK X 490.0mm. ID X 6096.0mm.	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 14.0mm. THK X 490.0mm. ID X 50.8mm.	SA-516 70
14	BAFFLES: 484.0mm DIA X 6.0mm THK	SA-516 70
1	TUBESHEET: 622.0mm X 52.0mm THK	SA-266 4
160	TUBES: 6093.0mm X 19.0mm DIA X 2.8mm THK	SA-179
1	INSULATION: 75mm X 45mm THK	HOT
2	SADDLE: 150.0mm X 120 DEG	SA-516 70
1	INSULATION: 6096mm X 45mm THK	HOT
1	INSULATION: 173mm X 45mm THK	HOT
1	GASKET: 518.0mm. OD X 498.0mm. ID	...
24	BODY FLANGE BOLTS: 19.0mm. DIA	SA-193 B7
48	NUTS FOR BODY FLANGE BOLTS: 19.0mm. DIA	...
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
Shell FLG [11]		B	-23	-47	-29	0.582	12.000	1.00	No
Main shell [8]		B	-23	-47	-29	0.577	12.000	1.00	No
Rear End Cove[10]		B	-23	-48	-29	0.562	12.000	1.00	No
Rear End Cover[7]		B	-18	-48	-29	0.470	14.000	1.00	No
Left Saddle [24]		B	-23	-47	-29	0.577	12.000	1.00	No
RIGHT SADDLE [24]		B	-23	-47	-29	0.577	12.000	1.00	No
SH IN NOZ S1 [1]		B	-23	-48	-29	0.354	12.000	1.00	No
Nozzle Flg [5]		A	-18	-104					
SH OUT NOZ S2 [1]		B	-23	-48	-29	0.355	12.000	1.00	No
Nozzle Flg [5]		A	-18	-104					
Tubesheet: SS[13]		B	-21	-27	-29	0.892	13.000	1.00	No
Warmest MDMT:			-18	-27					
CH FLG [11]		B	-23	-47	-29	0.582	12.000	1.00	No
Fr. END COVER[10]		B	-23	-48	-29	0.562	12.000	1.00	No
Fr. END COVER [7]		B	-18	-48	-29	0.470	14.000	1.00	No
CH.SHELL [8]		B	-23	-47	-29	0.577	12.000	1.00	No
TS IN NOZ T1 [1]		B	-8	-104	-29	0.056	9.601	1.00	No
Nozzle Flg [5]		A	-18	-104					
TS OUT NOZ T2 [1]		B	-8	-104	-29	0.056	9.601	1.00	No
Nozzle Flg [5]		A	-18	-104					
Tubesheet: CS[14]		B	-21	-27	-29	0.892	13.000	1.00	No
Bolting [21]			-48						
Warmest MDMT:			-8	-27					
Exchanger Side		Computed MDMT			Required MDMT		Pass/Fail		
Shell		-29.0			-28.9		Pass		
Channel/Tube		-29.0			-28.9		Pass		

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.



UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Warning: CH FLG and Shell FLG

There are mating flanges in this model that have different design pressures. This will generate different bolt loads. If a tubesheet is in between these flanges, PV Elite will compute the mating flange loads for you. If there is not a tubesheet physically separating these flanges, you must enter in the mating flange loads into the flange dialog.

Warning:

There is a flange in this model connected to an element whose temperature is different. The bolt allowable stress is based on it's parent design temperature. You may need to manually adjust the operating bolt allowable stress to a value based upon a representative maximum temperature the bolts will be subject to.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (Kgfcmm2mm):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	17.237	bars
Channel Side Design Pressure	5.516	bars
Shell Side Design Temperature	132.2	°C
Channel Side Design Temperature	76.7	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Sour	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-28.9	°C
MDMT (CET), Tube Side	-28.9	°C
User defined MAWP, Shell Side	0	bars
User defined MAWP, Channel Side	0	bars
User defined MAPnc, Shell Side	0	bars
User defined MAPnc, Channel Side	0	bars
User defined Test Pres., Shell Side	0	bars
User defined Test Pres., Channel Side	0	bars

Projection of Nozzle from Vessel Top	0	mm.
Projection of Nozzle from Vessel Bottom	0	mm.
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175



0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vortex Shedding N

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	Km/hr
Wind Zone Number	5	
Base Elevation	300	mm.
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	IS 1893-RSM
Importance Factor for IS-1893	2.000
Soil Factor	1.000
Zone Number	2.000
Percent Seismic for Hydrotest	0.000
Period of Vibration (User defined) Sec	0.000
Damping Factor	2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
-------------	------

0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vertical Acceleration Yes

Design Pressure + Static Head Y
 Consider MAP New and Cold in Noz. Design N
 Consider External Loads for Nozzle Des. Y
 Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No
 Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes
 Do not use Nozzle MDMT Interpretation VIII-1 01-37 No
 Use Table G instead of exact equation for "A" Yes
 Shell Head Joints are Tapered Yes
 Compute "K" in corroded condition Yes
 Use Code Case 2286 No
 Use the MAWP to compute the MDMT Yes
 For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes
 For PWHT & P1 Materials the MDMT can be < -55F (-48C) No

Using Metric Material Databases, ASME II D Yes
 Calculate B31.3 type stress for Nozzles with Loads Yes
 Reduce the MDMT due to lower membrane stress Yes
 Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/6
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	Fr. END COVER
Distance "FROM" to "TO"	50 mm.
Inside Diameter	490 mm.
Element Thickness	12 mm.
Internal Corrosion Allowance	3.175 mm.
Nominal Thickness	14 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	5.516 bars
Design Temperature Internal Pressure	77 °C
Design External Pressure	0 bars
Design Temperature External Pressure	77 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²



Allowable Stress, Hydrotest	179.4	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm.
Yield Stress, Operating	243.68	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid 10	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	490	mm.
Liquid Density	0.0009984	kg./cm ³

Element#	2/6	
Element From Node	20	
Element To Node	30	
Element Type	Cylinder	
Description	CH.SHELL	
Distance "FROM" to "TO"	400	mm.
Inside Diameter	490	mm.
Element Thickness	12	mm.
Internal Corrosion Allowance	3.175	mm.
Nominal Thickness	12	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	5.516	bars
Design Temperature Internal Pressure	77	°C
Design External Pressure	0	bars
Design Temperature External Pressure	77	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid: 20	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	490	mm.
Liquid Density	0.0009984	kg./cm ³

Element From Node	20	
Detail Type	Nozzle	
Detail ID	TS IN NOZ T1	
Dist. from "FROM" Node / Offset dist	200	mm.
Nozzle Diameter	6	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	-90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	31.17	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	TS OUT NOZ T2	
Dist. from "FROM" Node / Offset dist	200	mm.
Nozzle Diameter	6	in.
Nozzle Schedule	80	
Nozzle Class	300	



Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	31.17	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element#	3/6	
Element From Node	30	
Element To Node	40	
Element Type	Flange	
Description	CH FLG	
Distance "FROM" to "TO"	78	mm.
Flange Inside Diameter	490	mm.
Element Thickness	60	mm.
Internal Corrosion Allowance	3.175	mm.
Nominal Thickness	60	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	5.516	bars
Design Temperature Internal Pressure	77	°C
Design External Pressure	0	bars
Design Temperature External Pressure	77	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 4	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	179.4	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm.
Yield Stress, Operating	231	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03017	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid 30	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	490	mm.
Liquid Density	0.0009984	kg./cm ³

Element#	4/6	
Element From Node	40	
Element To Node	50	
Element Type	Flange	
Description	Shell FLG	
Distance "FROM" to "TO"	80	mm.
Flange Inside Diameter	490	mm.
Element Thickness	60	mm.
Internal Corrosion Allowance	3.175	mm.
Nominal Thickness	60	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	17.237	bars
Design Temperature Internal Pressure	132	°C
Design External Pressure	0	bars
Design Temperature External Pressure	132	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 4	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid 40	



Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	490	mm.
Liquid Density	0.0009984	kg./cm ³

Element From Node	40	
Detail Type	Insulation	
Detail ID	HOT INSULATION	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	75	mm.
Thickness of Insulation	45	mm.
Density	0.00015	kg./cm ³

Element#	5/6	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	Main shell	
Distance "FROM" to "TO"	6096	mm.
Inside Diameter	490	mm.
Element Thickness	12	mm.
Internal Corrosion Allowance	3.175	mm.
Nominal Thickness	12	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	17.237	bars
Design Temperature Internal Pressure	132	°C
Design External Pressure	0	bars
Design Temperature External Pressure	132	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	179.4	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm.
Yield Stress, Operating	234.13	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	1220	mm.
Width of Saddle	150	mm.
Height of Saddle at Bottom	500	mm.
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm.
Width of Wear Plate	225	mm.
Thickness of Wear Plate	12	mm.
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.44999999	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm.
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm.

Element From Node	50	
Detail Type	Saddle	
Detail ID	RIGHT SADDLE	
Dist. from "FROM" Node / Offset dist	4876	mm.
Width of Saddle	150	mm.
Height of Saddle at Bottom	500	mm.
Saddle Contact Angle	120.0	



Height of Composite Ring Stiffener	0	mm.
Width of Wear Plate	225	mm.
Thickness of Wear Plate	12	mm.
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.44999999	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm.
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm.

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid 50	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	490	mm.
Liquid Density	0.0009984	kg./cm ³

Element From Node	50	
Detail Type	Insulation	
Detail ID	HOT INSULATION	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	6096	mm.
Thickness of Insulation	45	mm.
Density	0.00015	kg./cm ³

Element From Node	50	
Detail Type	Nozzle	
Detail ID	SH IN NOZ S1	
Dist. from "FROM" Node / Offset dist	5846	mm.
Nozzle Diameter	6	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	33.677	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	SH OUT NOZ S2	
Dist. from "FROM" Node / Offset dist	250	mm.
Nozzle Diameter	8	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	-90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	46.732	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element#	6/6	
Element From Node	60	
Element To Node	70	
Element Type	Elliptical	
Description	Rear End Cover	
Distance "FROM" to "TO"	50.8	mm.
Inside Diameter	490	mm.
Element Thickness	12	mm.
Internal Corrosion Allowance	3.175	mm.
Nominal Thickness	14	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	17.237	bars
Design Temperature Internal Pressure	132	°C
Design External Pressure	0	bars
Design Temperature External Pressure	132	°C
Effective Diameter Multiplier	1.2	



Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	60
Detail Type	Liquid
Detail ID	Liquid 60
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	490 mm.
Liquid Density	0.0009984 kg./cm ³
Element From Node	60
Detail Type	Insulation
Detail ID	HOT INSULATION
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	173.3 mm.
Thickness of Insulation	45 mm.
Density	0.00015 kg./cm ³



XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
Fr. END COVER		50	...	50	...
CH.SHELL		450	...	400	...
CH FLG		532.5	...	78	...
Shell FLG		664.5	...	80	...
Main shell		6760.5	...	6096	...
Rear End Cover		6811.3	...	50.8	...

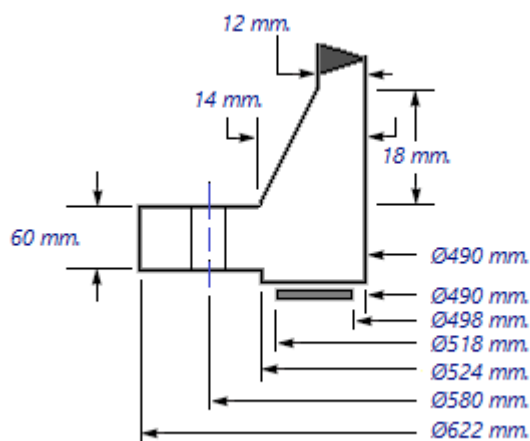
The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flange Input Data Values **Description: FLANGE** :

CH FLG

Flange Type	Integral Weld Neck		
Design Pressure	P	5.56	bars
Design Temperature		77	°C
Internal Corrosion Allowance	ci	3.1750	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	490.000	mm.
Flange Outside Diameter	A	622.000	mm.
Flange Thickness	t	60.0000	mm.
Thickness of Hub at Small End	go	12.0000	mm.
Thickness of Hub at Large End	gl	14.0000	mm.
Length of Hub	h	18.0000	mm.
Flange Material		SA-266 4	
Flange Material UNS number		K03017	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	580.000	mm.
Nominal Bolt Diameter	a	19.0500	mm.
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	524.000	mm.
Flange Face Inside Diameter	Fid	490.000	mm.
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	518.000	mm.
Gasket Inside Diameter	Gi	498.000	mm.
Gasket Factor	m	3.7500	
Gasket Design Seating Stress	y	62.06	N./mm ²
Gasket Thickness	tg	4.5000	mm.
Raised face depth	Fd	0.0000	mm.
Length of Partition Gasket	lp	498.0000	mm.
Width of Partition Gasket	tp	10.0000	mm.
Partition Gasket Factor	mPart	3.7500	
Partition Gasket Design Seating Stress	yPart	52.40	N./mm ²



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (5.56 \cdot (490.0/2 + 3.175)) / (138.0 \cdot 1.0 - 0.6 \cdot 5.56) + Ca$$

$$= 4.1781 \text{ mm.}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 8.825) / (248.175 + 0.6 \cdot 8.825)$$

$$= 48.043 \text{ bars}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (245.0 + 0.6 \cdot 12.0)$$

$$= 65.657 \text{ bars}$$

Corroded Flange ID,	Bcor = B+2*ci	496.350 mm.
Corroded Large Hub,	g1Cor = g1-ci	10.825 mm.
Corroded Small Hub,	g0Cor = go-ci	8.825 mm.
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	31.000 mm.
Gasket Contact Width,	N = (Go - Gi)/2	10.000 mm.
Basic Gasket Width,	bo = N/2	5.000 mm.
Effective Gasket Width,	b = bo	5.000 mm.
Gasket Reaction Diameter,	G = (Go + Gi)/2	508.000 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 508.0^2 \cdot 5.564$$

$$= 11500.146 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot lp \cdot bPart \cdot mPart \cdot P$$

$$= 2 \cdot 5.0 \cdot \pi \cdot 508.0 \cdot 3.75 \cdot 5.56$$

$$+ 2 \cdot 498.0 \cdot 5.0 \cdot 3.75 \cdot 5.564$$

$$= 4455.322 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 496.35^2 \cdot 5.564$$

$$= 10978.726 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 11500 - 10979$$

$$= 521.419 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(11500 + 4455 + 0, 0)$$

$$= 15955.468 \text{ Kg}$$

$$= 49566.719 \text{ Kg, Mating Flange Load Governs}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 62.06 \cdot 5.0 \cdot \pi \cdot 508.0 + 52.4 \cdot 5.0 \cdot 498.0$$

$$= 63798.156 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((49567 + 0 + 4 \cdot 0 / 508.0) / 172, 63798 / 172)$$

$$= 36.376 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 19.05 + 6 \cdot 60.0 / (3.75 + 0.5)$$

$$= 122.806 \text{ mm.}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 580.0 \cdot \sin(\pi / 24)$$

$$= 75.705 \text{ mm.}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{75.705 / (2 \cdot 19.05 + 60.0)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	36.376	46.761	
Radial Distance between Hub and Bolts:	28.575	31.000	
Radial Distance between Bolts and Edge:	20.637	21.000	
Circ. Spacing between the Bolts:	44.450	75.705	122.806

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 46.761 \cdot 172.0 / (62.06 \cdot \pi(518.0 + 498.0))$$

$$= 4.061 \text{ mm.}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 46.7612$$

$$= 82012.02 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 49567 - 11500$$

$$= 38066.57 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (580.0 - 508.0) / 2$$

$$= 36.0000 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((580.0 - 496.35) / 2 + 36.0)$$

$$= 38.9125 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (580.0 - 496.35 - 10.825) / 2$$

$$= 36.4125 \text{ mm.}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	10979.	36.4125	1.0000	400.
Face Pressure, Mt	521.	38.9125	1.0000	20.
Gasket Load, Mg	38067.	36.0000	1.0000	1370.
Gasket Seating, Matm	82012.	36.0000	1.0000	2952.

Total Moment for Operation, Mo 1790. Kg-m.
 Total Moment for Gasket seating, Mg 2952. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 66.184 mm.
 Hub Ratio, h/h0 = HL / H0 0.272
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.227

Flange Factors for Integral Flange:

Factor F per 4.16.5	0.899
Factor V per 4.16.5	0.448
Factor f per 4.16.5	1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 622.0/496.35$$

$$= 1.253$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.253^2 \cdot \log_{10}(1.253))) / (1.253^2 - 1)) / (1.253 - 1)$$

$$= 8.734$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.253^2(1 + 8.55246 \cdot \log_{10}(1.253)) - 1) / ((1.0472 + 1.9448 \cdot 1.253^2) \cdot (1.253 - 1))$$

$$= 1.817$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.253^2(1 + 8.55246 \cdot \log_{10}(1.253)) - 1) / (1.36136(1.253^2 - 1) \cdot (1.253 - 1))$$

$$= 9.598$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.253^2 + 1) / (1.253^2 - 1)$$

$$= 4.506$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 9.598 \cdot 8.825^2 \cdot 66.184 / 0.448$$

$$= 110540.781 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.899/66.184$$

$$= 0.014 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (60.0 \cdot 0.014 + 1) / 1.817 + 60.0^3/110540.781$$

$$= 2.953$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 1790/496.35) / (2.9529 \cdot 10.825^2)$$

$$= 102.24 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 2952/496.35) / (2.9529 \cdot 10.825^2)$$

$$= 168.59 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.0867 \cdot 1790/496.35) / (2.9529 \cdot 60.0^2)$$

$$= 6.94 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.0867 \cdot 2952/496.35) / (2.9529 \cdot 60.0^2)$$

$$= 11.45 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (8.734 \cdot 1790 / (60.0^2 \cdot 496.35)) - 4.5064 \cdot 7$$

$$= 54.53 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:
 $= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$
 $= (8.734 \cdot 2952 / (60.0^2 \cdot 496.35)) - 4.5064 \cdot 11$
 $= 89.92 \text{ N./mm}^2$

Average Flange Stress, Operating [SAo]:
 $= (SHo + \max(SRo, STo)) / 2$
 $= (102 + \max(7, 55)) / 2$
 $= 78.38 \text{ N./mm}^2$

Average Flange Stress, Seating [SAa]:
 $= (SHA + \max(SRa, STa)) / 2$
 $= (169 + \max(11, 90)) / 2$
 $= 129.25 \text{ N./mm}^2$

Bolt Stress, Operating [BSo]:
 $= Wm1 / Ab$
 $= 49567 / 46.7612$
 $= 103.95 \text{ N./mm}^2$

Bolt Stress, Seating [BSa]:
 $= (Wm2 / Ab)$
 $= (63798 / 46.7612)$
 $= 133.80 \text{ N./mm}^2$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	102.24	207.00	168.59	207.00
Radial Flange	6.94	138.00	11.45	138.00
Tangential Flange	54.53	138.00	89.92	138.00
Maximum Average	78.38	138.00	129.25	138.00
Bolting	103.95	172.00	133.80	172.00

Minimum Required Flange Thickness 57.810 mm.

Estimated M.A.W.P. (Operating) 29 bars

Estimated Finished Weight of Flange at given Thk. 56.5 kg.

Estimated Unfinished Weight of Forging at given Thk 69.7 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 2952.5 / 1.0 \cdot 9806.637 \cdot 0.448 / (2.729 \cdot 202713 \cdot 8.825^2 \cdot 66.184 \cdot 0.3)$
 $= 0.789 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 1790.5 / 1.0 \cdot 9806.637 \cdot 0.448 / (2.729 \cdot 199531 \cdot 8.825^2 \cdot 66.184 \cdot 0.3)$
 $= 0.486 \text{ (should be } \leq 1)$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 2952.5 / 1.0 \cdot 9806.637 \cdot 0.448 / (2.953 \cdot 202713 \cdot 8.825^2 \cdot 66.184 \cdot 0.3)$
 $= 0.729 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 1790.5 / 1.0 \cdot 9806.637 \cdot 0.448 / (2.953 \cdot 199531 \cdot 8.825^2 \cdot 66.184 \cdot 0.3)$
 $= 0.449 \text{ (should be } \leq 1)$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.582, Temperature Reduction per Fig. UCS 66.1 = 24 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
Min Metal Temp. at Required thickness (UCS 66.1) -47 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	93.91	207.00	153.23	207.00
Radial Flange	9.95	138.00	16.24	138.00
Tangential Flange	40.74	138.00	66.48	138.00
Maximum Average	67.32	138.00	109.85	138.00
Bolting	103.95	172.00	133.80	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 50.292 mm.

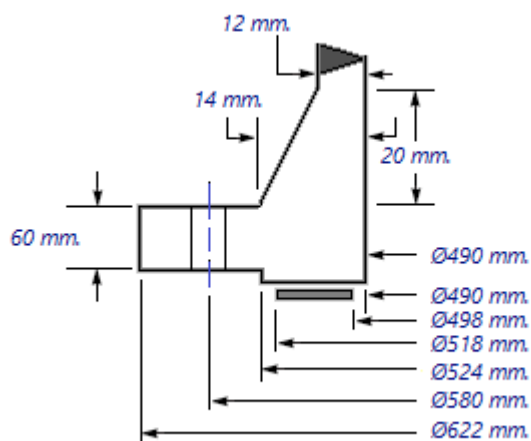
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Flange Input Data Values

Description: SH. FLG :

Shell FLG

Flange Type	Integral Weld Neck		
Design Pressure	P	17.29	bars
Design Temperature		132	°C
Internal Corrosion Allowance	ci	3.1750	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	490.000	mm.
Flange Outside Diameter	A	622.000	mm.
Flange Thickness	t	60.0000	mm.
Thickness of Hub at Small End	go	12.0000	mm.
Thickness of Hub at Large End	gl	14.0000	mm.
Length of Hub	h	20.0000	mm.
Flange Material		SA-266 4	
Flange Material UNS number		K03017	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	580.000	mm.
Nominal Bolt Diameter	a	19.0500	mm.
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	524.000	mm.
Flange Face Inside Diameter	Fid	490.000	mm.
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	518.000	mm.
Gasket Inside Diameter	Gi	498.000	mm.
Gasket Factor	m	3.7500	
Gasket Design Seating Stress	y	62.06	N./mm ²
Gasket Thickness	tg	4.5000	mm.
Raised face depth	Fd	0.0000	mm.
Length of Partition Gasket	lp	498.0000	mm.
Width of Partition Gasket	tp	10.0000	mm.
Partition Gasket Factor	mPart	3.7500	
Partition Gasket Design Seating Stress	yPart	52.40	N./mm ²





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Hub Small End Required Thickness due to Internal Pressure:

$$\begin{aligned} &= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= (17.29 \cdot (490.0/2 + 3.175)) / (138.0 \cdot 1.0 - 0.6 \cdot 17.29) + Ca \\ &= 6.3073 \text{ mm.} \end{aligned}$$

Small End Hub MAWP:

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\ &= (138.0 \cdot 1.0 \cdot 8.825) / (248.175 + 0.6 \cdot 8.825) \\ &= 48.043 \text{ bars} \end{aligned}$$

Hub Small End Hub MAPnc:

$$\begin{aligned} &= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\ &= (138.0 \cdot 1.0 \cdot 12.0) / (245.0 + 0.6 \cdot 12.0) \\ &= 65.657 \text{ bars} \end{aligned}$$

Corroded Flange ID,	Bcor = B+2*ci	496.350 mm.
Corroded Large Hub,	g1Cor = g1-ci	10.825 mm.
Corroded Small Hub,	g0Cor = go-ci	8.825 mm.
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	31.000 mm.
Gasket Contact Width,	N = (Go - Gi)/2	10.000 mm.
Basic Gasket Width,	bo = N/2	5.000 mm.
Effective Gasket Width,	b = bo	5.000 mm.
Gasket Reaction Diameter,	G = (Go + Gi)/2	508.000 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$\begin{aligned} &= \pi/4 \cdot G^2 \cdot P \\ &= \pi/4 \cdot 508.0^2 \cdot 17.285 \\ &= 35725.969 \text{ Kgf} \end{aligned}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$\begin{aligned} &= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot lp \cdot bPart \cdot mPart \cdot P \\ &= 2 \cdot 5.0 \cdot \pi \cdot 508.0 \cdot 3.75 \cdot 17.29 \\ &\quad + 2 \cdot 498.0 \cdot 5.0 \cdot 3.75 \cdot 17.285 \\ &= 13840.755 \text{ Kgf} \end{aligned}$$

Hydrostatic End Load at Flange ID [Hd]:

$$\begin{aligned} &= \pi/4 \cdot Bcor^2 \cdot P \\ &= \pi/4 \cdot 496.35^2 \cdot 17.285 \\ &= 34106.145 \text{ Kgf} \end{aligned}$$

Pressure Force on Flange Face [Ht]:

$$\begin{aligned} &= H - Hd \\ &= 35726 - 34106 \\ &= 1619.823 \text{ Kgf} \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned} &= \max(Wo,1 + Wo,2 + H'p, 0) \\ &= \max(35726 + 13841 + 0, 0) \\ &= 49566.719 \text{ Kgf} \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned} &= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp \\ &= 62.06 \cdot 5.0 \cdot \pi \cdot 508.0 + 52.4 \cdot 5.0 \cdot 498.0 \\ &= 63798.156 \text{ Kgf} \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned} &= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa) \\ &= \max((49567 + 0 + 4 \cdot 0 / 508.0) / 172, 63798 / 172) \\ &= 36.376 \text{ cm}^2 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$\begin{aligned} &= 2a + 6t/(m + 0.5) \\ &= 2 \cdot 19.05 + 6 \cdot 60.0/(3.75 + 0.5) \\ &= 122.806 \text{ mm.} \end{aligned}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 580.0 \cdot \sin(\pi / 24)$$

$$= 75.705 \text{ mm.}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{75.705 / (2 \cdot 19.05 + 60.0)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	36.376	46.761	
Radial Distance between Hub and Bolts:	28.575	31.000	
Radial Distance between Bolts and Edge:	20.637	21.000	
Circ. Spacing between the Bolts:	44.450	75.705	122.806

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 46.761 \cdot 172.0 / (62.06 \cdot \pi(518.0 + 498.0))$$

$$= 4.061 \text{ mm.}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 46.7612$$

$$= 82012.02 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 49567 - 35726$$

$$= 13840.75 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (580.0 - 508.0) / 2$$

$$= 36.0000 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((580.0 - 496.35) / 2 + 36.0)$$

$$= 38.9125 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (580.0 - 496.35 - 10.825) / 2$$

$$= 36.4125 \text{ mm.}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	34106.	36.4125	1.0000	1242.
Face Pressure, Mt	1620.	38.9125	1.0000	63.
Gasket Load, Mg	13841.	36.0000	1.0000	498.
Gasket Seating, Matm	82012.	36.0000	1.0000	2952.

Total Moment for Operation, Mo 1803. Kg-m.
 Total Moment for Gasket seating, Mg 2952. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 66.184 mm.
 Hub Ratio, h/h0 = HL / H0 0.302
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.227

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.895

Factor V per 4.16.5 0.441
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 622.0/496.35$$

$$= 1.253$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.253^2 \cdot \log_{10}(1.253))) / (1.253^2 - 1)) / (1.253 - 1)$$

$$= 8.734$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.253^2(1 + 8.55246 \cdot \log_{10}(1.253)) - 1) / ((1.0472 + 1.9448 \cdot 1.253^2) \cdot (1.253 - 1))$$

$$= 1.817$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.253^2(1 + 8.55246 \cdot \log_{10}(1.253)) - 1) / (1.36136(1.253^2 - 1) \cdot (1.253 - 1))$$

$$= 9.598$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.253^2 + 1) / (1.253^2 - 1)$$

$$= 4.506$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 9.598 \cdot 8.825^2 \cdot 66.184 / 0.441$$

$$= 112163.523 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.895/66.184$$

$$= 0.014 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (60.0 \cdot 0.014 + 1) / 1.817 + 60.0^3/112163.523$$

$$= 2.923$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 1803/496.35) / (2.9225 \cdot 10.825^2)$$

$$= 104.03 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 2952/496.35) / (2.9225 \cdot 10.825^2)$$

$$= 170.33 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.0817 \cdot 1803/496.35) / (2.9225 \cdot 60.0^2)$$

$$= 7.05 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.0817 \cdot 2952/496.35) / (2.9225 \cdot 60.0^2)$$

$$= 11.54 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (8.734 \cdot 1803 / (60.0^2 \cdot 496.35)) - 4.5064 \cdot 7$$

$$= 54.67 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:
 $= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$
 $= (8.734 \cdot 2952 / (60.0^2 \cdot 496.35)) - 4.5064 \cdot 12$
 $= 89.51 \text{ N./mm}^2$

Average Flange Stress, Operating [SAo]:
 $= (SHo + \max(SRo, STo)) / 2$
 $= (104 + \max(7, 55)) / 2$
 $= 79.35 \text{ N./mm}^2$

Average Flange Stress, Seating [SAa]:
 $= (SHA + \max(SRa, STa)) / 2$
 $= (170 + \max(12, 90)) / 2$
 $= 129.92 \text{ N./mm}^2$

Bolt Stress, Operating [BSo]:
 $= Wm1 / Ab$
 $= 49567 / 46.7612$
 $= 103.95 \text{ N./mm}^2$

Bolt Stress, Seating [BSa]:
 $= (Wm2 / Ab)$
 $= (63798 / 46.7612)$
 $= 133.80 \text{ N./mm}^2$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	104.03	207.00	170.33 207.00
Radial Flange	7.05	138.00	11.54 138.00
Tangential Flange	54.67	138.00	89.51 138.00
Maximum Average	79.35	138.00	129.92 138.00
Bolting	103.95	172.00	133.80 172.00

Minimum Required Flange Thickness 57.963 mm.

Estimated M.A.W.P. (Operating) 29 bars
 Estimated Finished Weight of Flange at given Thk. 56.8 kg.
 Estimated Unfinished Weight of Forging at given Thk 71.5 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 2952.5 / 1.0 \cdot 9806.637 \cdot 0.441 / (2.718 \cdot 202713 \cdot 8.825^2 \cdot 66.184 \cdot 0.3)$
 $= 0.781 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 1803.2 / 1.0 \cdot 9806.637 \cdot 0.441 / (2.718 \cdot 196164 \cdot 8.825^2 \cdot 66.184 \cdot 0.3)$
 $= 0.493 \text{ (should be } \leq 1)$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 2952.5 / 1.0 \cdot 9806.637 \cdot 0.441 / (2.923 \cdot 202713 \cdot 8.825^2 \cdot 66.184 \cdot 0.3)$
 $= 0.726 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 1803.2 / 1.0 \cdot 9806.637 \cdot 0.441 / (2.923 \cdot 196164 \cdot 8.825^2 \cdot 66.184 \cdot 0.3)$
 $= 0.458 \text{ (should be } \leq 1)$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.582, Temperature Reduction per Fig. UCS 66.1 = 24 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C



Min Metal Temp. at Required thickness (UCS 66.1) -47 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	97.23	207.00	154.17	207.00
Radial Flange	10.29	138.00	16.31	138.00
Tangential Flange	41.73	138.00	66.16	138.00
Maximum Average	69.48	138.00	110.17	138.00
Bolting	103.95	172.00	133.80	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 50.394 mm.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
Fr. END COVER		5.5643	14	3.175	490	138	1
CH.SHELL		5.5643	12	3.175	490	138	2
CH FLG		5.564	60	3.175	490	138	3
Shell FLG		17.285	60	3.175	490	138	4
Main shell		17.285	12	3.175	490	138	5
Rear End Cover		17.285	14	3.175	490	138	6

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
Fr. END COVER		5.516	49.6755	67.2573	12	5.675	1
CH.SHELL		5.516	47.9951	65.6572	12	5.675	2
CH FLG		5.516	28.5514	28.5994	60	57.8104	3
Shell FLG		17.237	28.5514	28.5994	60	57.9628	4
Main shell		17.237	47.9951	65.6572	12	6.30732	5
Rear End Cover		17.237	49.6755	67.2573	12	6.23558	6

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

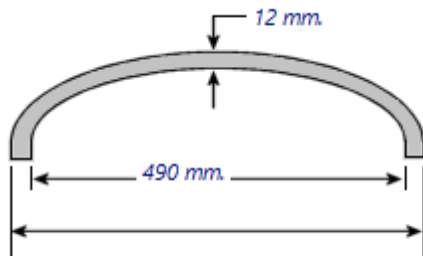
Shell Side MAWP = 28 bars
 Shell Side MAPnc = 29 bars
 Channel Side MAWP = 28 bars
 Channel Side MAPnc = 29 bars

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Fr. END COVER

Elliptical Head Design Information:					
Design Pressure	P	5.564 bars	Int. Design Temperature		77 °C
Material		SA-516 70	Ext. Design Temperature		77 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.1750 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	490.000 mm.	Actual Stress		15.4 N./mm ²
Spec. Min. Thk t,e, or sa		12.0000 mm.	Spec. Nominal Thk.		14.000 mm.
Surface Area		3722.3 cm ²	SG of Contents		1.000
Element Volume		24833.4 cm ³	Weight of Contents		24.8 Kg _f
Empty Weight		41.6 Kg _f	Operating Weight		66.4 Kg _f
			Comp. All. Ope. Stress		141.0 N./mm ²
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.125 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.062 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (5.564 \cdot 496.35 \cdot 0.983) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 5.564)$$

$$= 0.9844 + 3.1750 = 4.1594 \text{ mm.}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 2.5000 mm. per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.048 bars

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 8.825) / (0.983 \cdot 496.35 + 0.2 \cdot 8.825)$$

$$= 50 - 0 = 50 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 12.0) / (1.0 \cdot 490.0 + 0.2 \cdot 12.0)$$

$$= 67 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (5.564 \cdot (0.983 \cdot 496.35 + 0.2 \cdot 8.825)) / (2 \cdot 1.0 \cdot 8.825)$$

$$= 15.443 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (5.564 \cdot 248.175) / (138.0 \cdot 1.0 - 0.6 \cdot 5.564) + 3.175$$

$$= 4.178 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.048 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 10.825) / (248.175 + 0.6 \cdot 10.825)$$

$$= 59 - 0 = 59 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (496.35 / (2 \cdot 125.675)))^2 / 6$$

$$= 0.983264$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 11.628 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 12.0$, $t_r = 4.956$, $c = 3.175 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.562$, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 14.0$, $t_r = 5.092$, $c = 3.175 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.47$, Temp. Reduction = 37 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	27.970 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	12.000 mm.	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.562	Temperature Difference	25 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

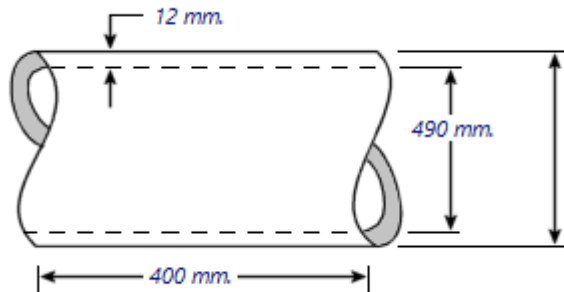
Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.125 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.062 mm.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: CH.SHELL

Cylindrical Shell Design Information:					
Design Pressure	P	5.564 bars	Int. Design Temperature		77 °C
Material		SA-516 70	Ext. Design Temperature		77 °C
Length		400.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.1750 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	490.000 mm.	Actual Stress		16.0 N./mm ²
Spec. Min. Thk t _e , or s _a		12.0000 mm.	Spec. Nominal Thk.		12.000 mm.
Surface Area		6459.1 cm ²	SG of Contents		1.000
Element Volume		75443.1 cm ³	Weight of Contents		75.4 Kg
Empty Weight		58.7 Kg	Operating Weight		134.1 Kg
			Comp. All. Ope. Stress		141.0 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (5.564 \cdot 248.175) / (138.0 \cdot 1.0 - 0.6 \cdot 5.564) \\
 &= 1.0032 + 3.1750 = 4.1782 \text{ mm.}
 \end{aligned}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 2.5000 mm. per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &\text{Less Operating Hydrostatic Head Pressure of 0.048 bars} \\
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 8.825) / (248.175 + 0.6 \cdot 8.825) \\
 &= 48 - 0 = 48 \text{ bars}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 12.0) / (245.0 + 0.6 \cdot 12.0) \\
 &= 66 \text{ bars}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (5.564 \cdot (248.175 + 0.6 \cdot 8.825)) / (1.0 \cdot 8.825) \\
 &= 15.983 \text{ N./mm}^2
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 2.390 %



Minimum Design Metal Temperature Results:

Govrn. thk, tg = 12.0, tr = 5.092, c = 3.175 mm., $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.577$, Temp. Reduction = 24 °C

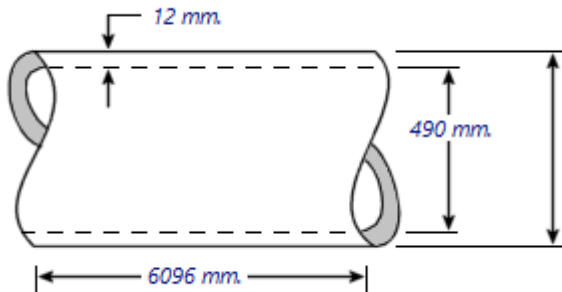
Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -47 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	27.970 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	12.000 mm.	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.577	Temperature Difference	24 °C
UCS-66(c) red.	No	Computed Min. Temp.	-47 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Main shell

Cylindrical Shell Design Information:					
Design Pressure	P	17.285 bars	Int. Design Temperature		132 °C
Material		SA-516 70	Ext. Design Temperature		132 °C
Length		6096.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.1750 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	490.000 mm.	Actual Stress		49.6 N./mm ²
Spec. Min. Thk t _e , or s _a		12.0000 mm.	Spec. Nominal Thk.		12.000 mm.
Surface Area		98436.9 cm ²	SG of Contents		1.000
Element Volume		860224.4 cm ³	Weight of Contents		859.7 Kg
Empty Weight		894.1 Kg	Operating Weight		1753.8 Kg
			Comp. All. Ope. Stress		141.0 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (17.285 \cdot 248.175) / (138.0 \cdot 1.0 - 0.6 \cdot 17.285)$$

$$= 3.1323 + 3.1750 = 6.3073 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.048 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 8.825) / (248.175 + 0.6 \cdot 8.825)$$

$$= 48 - 0 = 48 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (245.0 + 0.6 \cdot 12.0)$$

$$= 66 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (17.285 \cdot (248.175 + 0.6 \cdot 8.825)) / (1.0 \cdot 8.825)$$

$$= 49.650 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 2.390 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 12.0, t_r = 5.092, c = 3.175 mm., E* = 1.0



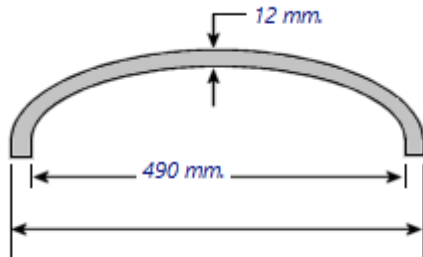
Thickness Ratio = $tr * E^* / (tg - c) = 0.577$, Temp. Reduction = 24 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
Min Metal Temp. at Required thickness (UCS 66.1) -47 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	27.970 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	12.000 mm.	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.577	Temperature Difference	24 °C
UCS-66(c) red.	No	Computed Min. Temp.	-47 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Rear End Cover

Elliptical Head Design Information:					
Design Pressure	P	17.285 bars	Int. Design Temperature		132 °C
Material		SA-516 70	Ext. Design Temperature		132 °C
Length		50.800 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.1750 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	490.000 mm.	Actual Stress		48.0 N./mm ²
Spec. Min. Thk t _e , or s _a		12.0000 mm.	Spec. Nominal Thk.		14.000 mm.
Surface Area		3735.3 cm ²	SG of Contents		1.000
Element Volume		24984.3 cm ³	Weight of Contents		25.0 Kg _f
Empty Weight		41.7 Kg _f	Operating Weight		66.7 Kg _f
			Comp. All. Ope. Stress		141.0 N./mm ²
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.125 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.062 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (17.285 \cdot 496.35 \cdot 0.983) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 17.285)$$

$$= 3.0606 + 3.1750 = 6.2356 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.048 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 8.825) / (0.983 \cdot 496.35 + 0.2 \cdot 8.825)$$

$$= 50 - 0 = 50 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 12.0) / (1.0 \cdot 490.0 + 0.2 \cdot 12.0)$$

$$= 67 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (17.285 \cdot (0.983 \cdot 496.35 + 0.2 \cdot 8.825)) / (2 \cdot 1.0 \cdot 8.825)$$

$$= 47.972 \text{ N./mm}^2$$

Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (17.285 \cdot 248.175) / (138.0 \cdot 1.0 - 0.6 \cdot 17.285) + 3.175$
 $= 6.307 \text{ mm.}$

Straight Flange Maximum Allowable Working Pressure:
 Less Operating Hydrostatic Head Pressure of 0.048 bars
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138.0 \cdot 1.0 \cdot 10.825) / (248.175 + 0.6 \cdot 10.825)$
 $= 59 - 0 = 59 \text{ bars}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (496.35 / (2 \cdot 125.675))^2) / 6$
 $= 0.983264$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 11.628 %
 Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 12.0$, $t_r = 4.956$, $c = 3.175 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.562$, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 14.0$, $t_r = 5.092$, $c = 3.175 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.47$, Temp. Reduction = 37 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	27.970 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	12.000 mm.	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.562	Temperature Difference	25 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.125 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.062 mm.

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	36.361 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	22.408 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	37.130 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	30.767 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	24.649 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	24.649 bars
App 27-4	= M.A.W.P.	27.970 bars

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	36.361 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	7.171 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	37.131 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	30.767 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	7.888 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	7.888 bars
App 27-4	= M.A.W.P.	27.970 bars

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Fr. END COVER	101.0	179.4	0.563	36.41
CH.SHELL	104.6	179.4	0.583	36.41
Main shell	104.6	179.4	0.583	36.41
Rear End Cover	101.0	179.4	0.563	36.41

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
TS IN NOZ T1	Nozzle	118.00	118.00	1.000
TS IN NOZ T1	Pad	138.00	138.00	1.000
TS OUT NOZ T2	Nozzle	118.00	118.00	1.000
TS OUT NOZ T2	Pad	138.00	138.00	1.000
SH IN NOZ S1	Nozzle	118.00	118.00	1.000
SH OUT NOZ S2	Nozzle	118.00	118.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Fr. END COVER	138.00	138.00	1.000
CH.SHELL	138.00	138.00	1.000
CH FLG	138.00	138.00	1.000
Shell FLG	138.00	138.00	1.000
Main shell	138.00	138.00	1.000
Rear End Cover	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	92.40	92.40	1.000



Tubesheet Material	138.00	138.00	1.000
--------------------	--------	--------	-------

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
TS IN NOZ T1	37.83	153.40	0.247
TS OUT NOZ T2	37.83	153.40	0.247
SH IN NOZ S1	37.83	153.40	0.247
SH OUT NOZ S2	40.42	153.40	0.263

Elements Suitable for Test Pressure.

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Input Echo, Tubesheet Number 1, Description: U-Tubesheet

Shell Data:

Main Shell Description: Main shell

Shell Test Pressure		36.36 bars
Shell Thickness	ts	12.0000 mm.
Shell Internal Corrosion Allowance	cas	0.0000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	490.000 mm.
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: CH.SHELL

Channel Type:		Cylinder
Channel Test Pressure		36.36 bars
Channel Thickness	tc	12.0000 mm.
Channel Internal Corrosion Allowance	cac	0.0000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	490.000 mm.
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	160
Tube Wall Thickness	et	2.7700 mm.
Tube Outside Diameter	D	19.0500 mm.
Total Straight Tube Length	Lt	6093.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		6041.00 mm.
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	180.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm.
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.7700 mm.
Groove Weld Leg	ag	2.7700 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	228.665 mm.
Largest Center-to-Center Tube Distance	Ul	35.0000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	6.0000 mm.

Tubesheet Data:

Tubesheet TYPE: U-tube, Gasketed both Sides, Conf. d

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-266 4



See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03017
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	52.0000 mm.
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm.
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm.
Tubesheet Outside Diameter	A	622.000 mm.

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		No
Step 1 Diameter on the Tubeside	dt1	521.00 mm.
Step 1 Depth on the Tubeside	ht1	6.00 mm.
Step 2 Diameter on the Tubeside	dt2	487.00 mm.
Step 2 Depth on the Tubeside	ht2	6.00 mm.
Is the Tubesheet Flat on Shellside?		No
Step 1 Diameter on the Shellside	ds1	521.00 mm.
Step 1 Depth on the Shellside	hs1	6.00 mm.
Step 2 Diameter on the Shellside	ds2	487.00 mm.
Step 2 Depth on the Shellside	hs2	6.00 mm.
Calculated Tubesheet Diameter per Div 2, 4.18.4		487.00 mm.

Note: Tubesheet diameter is now: 487.000 mm. Div 2, 4.18.4.

Dimension G for the Channel Side	Gc	508.000 mm.
Area of the Untubed Lanes	AL	253.0 cm ²

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Both
Flange Outside Diameter	A	622.000 mm.
Flange Inside Diameter	B	490.000 mm.
Flange Face Outside Diameter	Fod	524.000 mm.
Flange Face Inside Diameter	Fid	490.000 mm.
Gasket Outside Diameter	Go	518.000 mm.
Gasket Inside Diameter	Gi	498.000 mm.
Small end Hub thk.	g0	12.0000 mm.
Large end Hub thk.	g1	14.0000 mm.
Gasket Factor,	m	3.75
Gasket Design Seating Stress	y	62.06 N./mm ²
Flange Facing Sketch	Code Sketch 1a	
Gasket Thickness	tg	4.5000 mm.
Full face Gasket Flange Option	Program Selects	
Length of Partition Gasket	lp	498.000 mm.
Width of Partition Gasket	wp	10.0000 mm.
Partition Gasket Factor,	mPart	3.7500
Partition Gasket Design Seating Stress	yPart	52.40 N./mm ²
Partition Gasket Facing Sketch	Code Sketch 1a	
Partition Gasket Column for Gasket Seating	Code Column II	

Bolting Information:

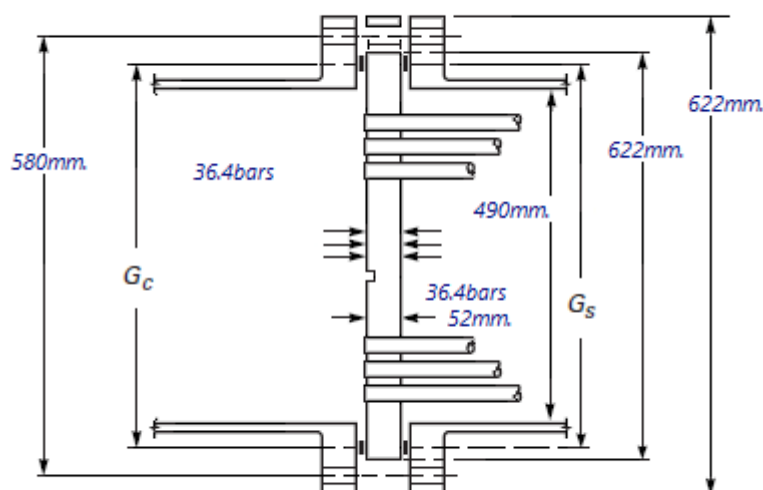
Diameter of Bolt Circle	C	580.000 mm.
Nominal Bolt Diameter	dB	19.0500 mm.
Thread Series	TEMA Thread Series	
Number of Bolts	n	24
Bolt Material	SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm.

Alternate Flange Operating Bolt Load, Wm1	104268.09 Kg
Alternate Flange Seating Bolt Load, Wm2	63798.16 Kg
Alternate Flange Design Bolt Load, W	93140.06 Kg



Tubesheet Integral with	None
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet Tf	32.0000 mm.
Is Bolt Load Transferred to the Tubesheet	Yes

Tubesheet Gasketed w/Shell and Channel



Configuration d:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (36.36 \cdot (19.05/2 - 0)) / (92.4 \cdot 1.0 + 0.4 \cdot 36.36)$
 $= 0.3690 + 0.0000 = 0.3690 \text{ mm.}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 21.11 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):



TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.7700	19.05	6269.67	6.88	50.0000	0.0232575	95.84

EMAWP = $(2.167 / (D/T) - 0.0833) * B = 222.1421$ bars

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.6458	19.05	6269.67	29.50	50.0000	0.0012643	80.45

EMAWP = $(4*B) / (3*(D/T)) = (4 * 80.4497) / (3 * 29.4962) = 36.364$ bars

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.6458 mm.
Allowable Internal Pressure at Corroded thickness	304.06 bars
Required Internal Design Pressure	36.36 bars
Allowable External Pressure at Corroded thickness	222.14 bars
Required External Design Pressure	36.36 bars
Required Thickness due to Shell Side pressure	0.6458 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

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Gasket Contact Width,	$N = (Goc - Gic) / 2$	10.000 mm.
Basic Gasket Width,	$b0 = N / 2$	5.000 mm.
Effective Gasket Width,	$b = b0$	5.000 mm.
Gasket Reaction Diameter,	$G = (Go + Gi) / 2$	508.000 mm.

Warning: Available bolt area, 46.761 cm² for then shell flange is < required 59.451 cm²

Note that for the pressure test case, ASME VIII-2 paragraph 4.18.13 (c)(9) does not require that the bolt area requirement be met.

Results for ASME U-tube Tubesheet Calculations for Configuration d:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:
 $= 2 * ro + dt$
 $= 2 * 228.665 + 19.05$
 $= 476.380$ mm.

Basic ligament efficiency for shear [μ]:
 $= (p - dt) / p$
 $= (31.75 - 19.05) / 31.75$
 $= 0.400$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:
 $= Gs / Do$ (Configurations d, e, f)
 $= 508.0 / 476.38$
 $= 1.0664$

Ratio [pc]:
 $= Gc / Do$ (Configurations d)
 $= 508.0 / 476.38$
 $= 1.0664$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:
 $= ltx / h$
 $= 0 / 52.0$
 $= 0.000$ (must be $0 \leq p \leq 1$)



Effective tube hole Diameter [d*]:

$$= \max(dt - 2tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2tt)$$

$$= \max(19.05 - 2 \cdot 2.77 (0.20270E+09/.20270E+09) \cdot (92/137) \cdot (0); 19.05 - 2 \cdot 2.77)$$

$$= 19.0500 \text{ mm.}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$$

$$= 31.75 / \sqrt{ (1 - 4 \cdot \min(252.98 ; 4 \cdot 476.38 \cdot 31.75) / (\pi \cdot 476.38^2)) }$$

$$= 34.2755 \text{ mm.}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (34.2755 - 19.05) / 34.2755$$

$$= 0.4442$$

$h/p = 1.637795$; $\mu^* = 0.444209$; $E^* = 98626944 \text{ KPa.}$

$E^*/E = 0.486563$; $v^* = 0.297378$

Skip Step 4 for Configuration d :

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / Do$$

$$= 487.0/476.38$$

$$= 1.0223$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (E \cdot \ln(K))$$

$$= (1 - 0.3) / 98626944 \cdot (0.20270E+09 \cdot \ln(1.02))$$

$$= 0.0318$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt]$$

$$= 476.38^2/16 \cdot [(1.066 - 1) \cdot (1.066^2 + 1) \cdot 36.361 - (1.066 - 1) \cdot (1.066^2 + 1) \cdot 36.361]$$

$$= 0.0000 \text{ bars} \cdot \text{mm.}^2$$

Moment acting on unperforated tubesheet rim [M*]

$$= Mts + W^* \cdot (Gc - Gs) / (2 \cdot \pi \cdot Do)$$

$$= 0 + 104268.1 \cdot (508.0 - 508.0) / (2 \cdot \pi \cdot 476.38)$$

$$= 0.0000 \text{ bars} \cdot \text{mm.}^2$$

Note: W is the maximum of the bolt loads between the shell and channel sides.*

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M^*) - Do^2/32 \cdot F(Ps - Pt)) / (1 + F)$$

$$= ((0) - 476.38^2/32 \cdot 0.032 (36.36 - 36.36)) / (1 + 0.03)$$

$$= 0.0000 \text{ bars} \cdot \text{mm.}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot (3 + v^*) \cdot (Ps - Pt)$$

$$= 0 + 476.38^2/64 \cdot (3 + 0.297) \cdot (36.36 - 36.36)$$

$$= 0.0000 \text{ bars} \cdot \text{mm.}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| Mp |; | Mo |)$$

$$= \max(| 0 |; | 0 |)$$

$$= 0.0000 \text{ bars} \cdot \text{mm.}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - hg')^2)$$



$$= 6 \cdot 0 / (0.4442 (52.0 - 6.0)^2) \\ = 0.0000 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot \beta T \cdot S_y \\ = 2 \cdot 0.95 \cdot 250.0 \\ = 474.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ = 6 \cdot 0 / (0.4442 (0.508 - 6.0)^2) \\ = 0.0000 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc} \\ = 0.508 + 0 + 0 \\ = 0.5080 \text{ mm.}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS}) \\ = \max(0.508; 0) \\ = 0.5080 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$= | P_s - P_t | \\ = | 36.36 - 36.36 | \\ = 0.000 \text{ bars}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / D_o \cdot 0.8 \cdot \beta T \cdot S_y \\ = 4 \cdot 0.4 \cdot 52.0 / 476.38 \cdot 0.8 \cdot 0.95 \cdot 249.995 \\ = 33.183 \text{ N./mm}^2$$

| Ps - Pt | is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu) \cdot (D_o / (0.8 \cdot \beta T \cdot S_y)) \cdot | P_s - P_t | + C_{ats} + C_{atc} \\ = 1 / (4 \cdot 0.4) \cdot (476.38 / 189.997) \cdot | 36.36 - 36.36 | + 0 \\ = 0.0000 \text{ mm.}$$

Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [At]:

$$= \pi (d_o - t) t \\ = \pi (19.05 - 2.77) 2.77 \\ = 1.417 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S \\ = \pi \cdot 2.77 (19.05 - 2.77) 170.997 \\ = 2470.266 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t) \\ = 0.85 \cdot \pi \cdot 2.77 (19.05 + 0.67 \cdot 2.77) 171.0 \\ = 2470.2656 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\ = 170.997 / 170.997 \\ = 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 2.77 (19.05 + 0.67 \cdot 2.77) 171.0$$

$$= 1744.7003 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd)} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 19.05)^2 + 1.07 \cdot 2.77 \cdot (19.05 - 2.77) \cdot 1.0 \cdot 1.0)} - 0.75 \cdot 19.05 \right)$$

$$= 3.198 \text{ mm.}$$

Minimum Required Fillet Weld Leg [afr]: 1.6620 mm.

Minimum Required Groove Weld Leg [agr]: 1.6620 mm.

Tube-Tubesheet Jt allowable, 2470.27 is >= tube strength 2470.27 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	0.0 <=	475.0 N./mm ²	Ok
Tubesheet shear stress	0.0 <=	190.0 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	0.508	52.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.662	2.770	Ok
Tube-Tubesheet Groove Weld Leg :	1.662	2.770	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/Fail
D1uc	43.244 ...	312	475 21 190	Ps+Pt D1	Ok
D2uc	43.244 24.077	312	475 21 190	Ps+Pt D2	Ok
D3uc	0.508 ...	...	475 ... 190	Ps+Pt D3	Ok
Max:	43.2435 24.077 mm.	0.656	0.110 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:



Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	43.244	52.000	Ok
Tubesheet thickness flanged extension :	24.077	32.000	Ok
Tube thickness :	0.646	2.770	Ok
Tube-tubesheet fillet weld leg :	1.662	2.770	Ok
Tube-tubesheet groove weld leg :	1.662	2.770	Ok

Note: This is a full strength Tube to Tubesheet Joint.

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External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	514	8.825	0.0023846	107.531
20	30	490.833	514	8.825	0.0032165	113.11
30	40	No Calc	...	56.825	No Calc	No Calc
40	50	No Calc	...	56.825	No Calc	No Calc
50	60	6187.63	514	8.825	0.0003364	33.6329
60	70	No Calc	514	8.825	0.0023846	107.531

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	12	No Calc	...	20.5125
20	30	12	No Calc	...	25.892
30	40	60	57.8104	...	No Calc
40	50	60	57.9628	...	No Calc
50	60	12	No Calc	...	7.69892
60	70	12	No Calc	...	20.5125

Minimum

8

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	490.833	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	6187.63	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	41.5786	24833.4	32.1492	25686	4.15786
20	30	58.6711	75443.1	43.4206	77411.2	5.86711
30	40	56.4764	15006.6	53.5026	15095.2	5.64764
40	50	56.7948	15383.8	53.7448	15482.2	5.67948
50	60	894.148	860224	661.729	890217	89.4148
60	70	41.716	24984.3	32.2554	25840.8	4.1716
Total		1149	1015875.62	876	1049732.75	114



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
10	Liqd	24.8182	-40.8333	...	...	Liquid 10
20	Liqd	75.3971	200	...	...	Liquid: 20
20	Nozl	34.2868	200	-318.165	...	TS IN NOZ T1
20	Nozl	34.2868	200	318.165	...	TS OUT NOZ T2
30	Liqd	14.9975	39	...	...	Liquid 30
40	Liqd	15.3744	40	...	...	Liquid 40
40	Insl	1.08121	37.5	...	...	HOT INSULATION
50	Sadl	49.723	1220	360.5	...	Left Saddle
50	Sadl	49.723	4876	360.5	...	RIGHT SADDLE
50	Liqd	859.7	3048	...	...	Liquid 50
50	Insl	72.3467	3048	...	...	HOT INSULATION
50	Nozl	37.0446	5846	318.165	...	SH IN NOZ S1
50	Nozl	51.4053	250	-341.837	...	SH OUT NOZ S2
60	Liqd	24.969	91.6333	...	...	Liquid 60
60	Insl	3.3753	86.65	...	...	HOT INSULATION
30	FTsh	114.492	108.5	...	...	U-Tubesheet
30	Tube	1224.9	3129	...	...	Tubes
40	Baf1	5.22512	466	...	...	Baffles
40	Baf1	5.22512	891	...	...	Baffles
40	Baf1	5.22512	1316	...	...	Baffles
40	Baf1	5.22512	1741	...	...	Baffles
40	Baf1	5.22512	2166	...	...	Baffles
40	Baf1	5.22512	2591	...	...	Baffles
40	Baf1	5.22512	3016	...	...	Baffles
40	Baf1	5.22512	3441	...	...	Baffles
40	Baf1	5.22512	3866	...	...	Baffles
40	Baf1	5.22512	4291	...	...	Baffles
40	Baf1	5.22512	4716	...	...	Baffles
40	Baf1	5.22512	5141	...	...	Baffles
40	Baf1	5.22512	5566	...	...	Baffles
40	Baf1	5.22512	5991	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	99.4
Liquid	1015.3
Insulation	76.8
Nozzles	157.0
Exchanger Components	1412.5
Liquid in Tubes	147.0
Sum of the Detail Weights	2908.1 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	1264.3	1264.3	1264.3	1264.3	1264.3	1264.3
Saddles	99.4	99.4	99.4	99.4	99.4	99.4
Nozzles	157.0	157.0	157.0	157.0	157.0	157.0
Exchanger	1412.5	1412.5	1412.5	1412.5	1412.5	1412.5
Insulation	...	...	76.8	76.8	76.8	76.8
Ope. Liquid	...	...	...	...	...	1015.3
Tube Ope Lqd	...	...	...	...	...	147.0
Test Liquid	...	1015.3	...	...	...	...
Tube Tst Lqd	...	148.6	...	...	...	...
Totals	2933.3	4097.2	3010.1	3010.1	3010.1	4172.4

Field Installation Options:

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	2933.3 kg.
Shop Test Wt.	- Fabricated weight + water (full)	4097.2 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	3010.1 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	3010.1 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	3010.1 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	4172.4 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	3872.6 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	148645.5 cm ³
Weight of Ope Liq in tubes :	147.0 kg.
Weight of Water in tubes :	148.6 kg.

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	3722.26	3372.34
20	30	6459.11	6157.52
30	40	2618.2	1200.72
40	50	2650.75	1231.5
50	60	98436.9	93840.6
60	70	3735.28	3384.65

Total	117622.5	109187.4 cm ²
Total	126.6	117.5 ft ²

Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
TS IN NOZ T1	48.15	51.00	77	300	GR 1.1	...	...	...	...
TS OUT NOZ T2	48.15	51.00	77	300	GR 1.1	...	...	...	...
SH IN NOZ S1	45.67	51.00	132	300	GR 1.1	31.710	36.794	45.669	36.794
SH OUT NOZ S2	45.67	51.00	132	300	GR 1.1	27.642	40.862	45.669	40.862

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 36.794 bars
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 51.000 bars

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 48.147 bars
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 51.000 bars

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
SH IN NOZ S1	196.088	0.50	459	408	27.970	36.794	Pass
SH OUT NOZ S2	248.539	0.50	816	551	27.970	40.862	Pass

Evaluating Flange on Nozzle [SH IN NOZ S1]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(459) + 4 \cdot \text{abs}(408) \cdot 196.088 \leq \pi \cdot 196.088^3 \cdot [(45.669 - 27.970) + .500 \cdot 45.669]$$

$$= 0.76596E+04 \leq 0.97911E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle SH IN NOZ S1 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(45.669, 0.500 \cdot 45.669 + 45.669 - 7659.6 / (\pi \cdot 196.088^3))$$

$$= \min(45.669, 36.794)$$

$$= 36.794 \text{ bars}$$

Evaluating Flange on Nozzle [SH OUT NOZ S2]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(816) + 4 \cdot \text{abs}(551) \cdot 248.539 \leq \pi \cdot 248.539^3 \cdot [(45.669 - 27.970) + .500 \cdot 45.669]$$

$$= 0.13596E+05 \leq 0.19937E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle SH OUT NOZ S2 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(45.669, 0.500 \cdot 45.669 + 45.669 - 13595.7 / (\pi \cdot 248.539^3))$$

$$= \min(45.669, 40.862)$$

$$= 40.862 \text{ bars}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020
Basic Wind Speed for IS-875	180 Km/hr
Wind Zone Number	5
Base Elevation	300 mm.
Percent Wind for Hydrotest	33.0
Risk Factor	[k1] 1.0
Terrain Category	2
Equipment Class	Class A
Topography Factor	[k3] 1.0
Importance Factor	[k4] 1.15
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Use Gust Response Factor (Dynamic Analysis)	No

Calculated Values:

Vessel Length to Diameter ratio: 11.821

Design Wind Speed [Vz]:
 = Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)
 = 50.0 • 1.0 • 1.0 • 1.0 • 1.15
 = 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:
 = max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)
 = max(0.7 • 1983.503, 1.0 • 1.0 • 1.0 • 0.6 • 57.5²)
 = 1983.5 N/m²

Element Wind Load (operating condition):
 = Wind Area • Cf • pd
 = 0.1 • 0.918 • 1983.503
 = 155.1 N

From	Height mm.	k1	k2	k3	Vz m/sec	Cf
10	800.00	1.0000	1.0000	1.0000	57.50	0.9182
20	800.00	1.0000	1.0000	1.0000	57.50	0.9182
30	800.00	1.0000	1.0000	1.0000	57.50	0.9182
40	800.00	1.0000	1.0000	1.0000	57.50	0.9182
50	800.00	1.0000	1.0000	1.0000	57.50	0.9182
60	800.00	1.0000	1.0000	1.0000	57.50	0.9182

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm²	Wind Pressure Kgs/m²	Element Wind Load Kgf
10	20	800	616.8	851.369	202.347	17.3938
20	30	800	616.8	2467.2	202.347	50.4057
30	40	800	588	458.64	202.347	9.37017
40	50	800	689.25	551.4	202.347	11.2653
50	60	800	724.8	44183.8	202.347	902.69
60	70	800	724.8	1219.71	202.347	24.9191

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

PV Elite 28 Licensee: SL User
FileName : 403-E-003
Wind Load Calculation:

Step: 14 2:18pm

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HEXAGON

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Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User I 2.0000
Zone Number 2

Soil Type Rocky or Hard Soil
Response reduction factor (Force factor) R 2.0000
Time Period of Vibration (program or user) 0.0303 Seconds
Value of Sa/g from user Spectrum Data 0.0939
Percent Damping 2.0 %
Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
= (Sa/g)ss • DampVal) • (I / R)
= (0.0939 • 1.0) • 1.0
= 0.094

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.0303 secs
Computed Sa/g = 0.0939

Note:

It was indicated to ignore the damping factor, assuming that the Site Spectrum specified is per the actual damping value of this vessel.
The value of "DampVal" has been set to 1 (not used).

Earthquake Element Load, for the first Element:
= Earthquake Weight • Ah • Scalar
= 471.64 • 0.0939 • 1.0
= 44.3 Kgf

Vertical load per 8.4:
= 2/3 • Ah • W • Scalar
= 2/3 • 0.0939 • 3773 • 1.0
= 236.1 Kgf

Earthquake Load Calculation:

Earthquake	Earthquake	Element
------------	------------	---------



From	To	Height mm.	Weight Kgf	Ope Load Kgf
10	20	245	471.64	44.2663
20	30	245	471.64	44.2663
30	40	245	471.64	44.2663
40	50	245	471.64	44.2663
50	Sad1	245	471.64	44.2663
Sad1	60	245	471.64	44.2663
50	60	245	471.64	44.2663
60	70	245	471.64	44.2663

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	3712.500 mm.
Center of Gravity of the Liquid	3346.510 mm.
Center of Gravity of the Insulation	3806.752 mm.
Center of Gravity of the Nozzles	1944.501 mm.
Center of Gravity of the Tubesheet(s)	558.500 mm.
Center of Gravity of the Tubes	3579.000 mm.
Center of Gravity of the Baffles	3813.000 mm.
Center of Gravity of Bare Shell New and Cold	3203.154 mm.
Center of Gravity of Bare Shell Corroded	3133.185 mm.
Vessel CG in the Operating Condition	3263.806 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	3236.896 mm.
Vessel CG in the Test Condition	3275.731 mm.

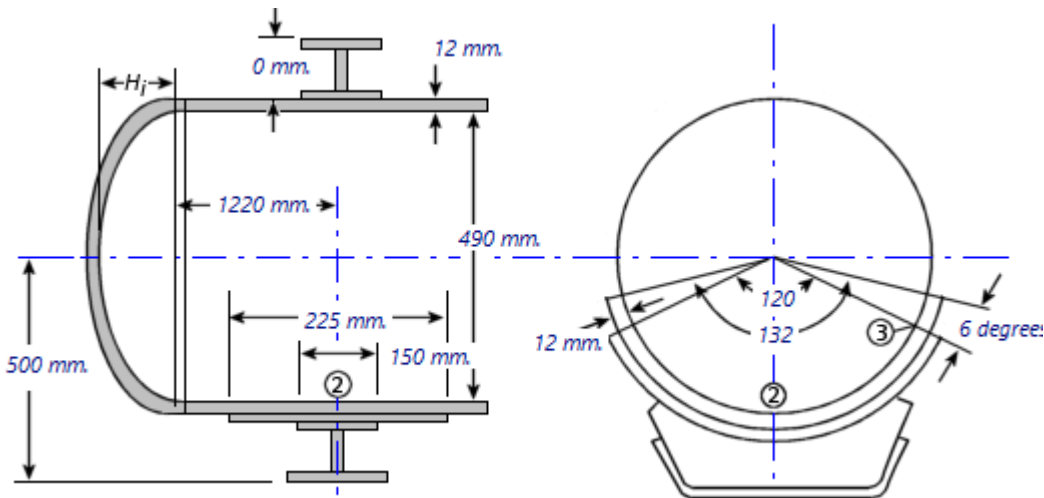
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	252.59	mm.
Shell Thickness used in this Case	t	12.000	mm.
Stiffened Vessel Length per 4.15.6	L	6226.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	1220.00	mm.
Saddle Width	b1 or b	150.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	225.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	500.00	mm.
Coefficient of Friction	mu	0.45	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		4136.47	Kgf
Pressure used in Analysis	P	17.261	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	16.86	138.00
Long. Stress at Bottom of Midspan	32.54	138.00
Long. Stress at Top of Saddles	75.02	138.00
Long. Stress at Bottom of Saddles	-3.19	-210.72
Long. Stress at Bottom of Saddles	3.19	138.00
Tangential Shear in Shell	12.96	110.40
Circ. Stress at Horn of Saddle	9.59	172.50
Circ. Compressive Stress in Shell	0.66	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:



Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(1016.0/2 + 0) \cdot 500.0/460.0$$

$$= 1656.6 \text{ Kg}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(66.51, 0) \cdot 500.0/3656.0007$$

$$= 9.1 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(354.13, 0) \cdot 500.0/3656.0007$$

$$= 48.4 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(354/2 + 0) \cdot 500.0/460.0$$

$$= 577.4 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 2480 + \max(9, 1657, 48, 577)$$

$$= 4136.5 \text{ Kg}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1984.306(0.0 + \pi/4(0.514 \cdot 1.2)^2)$$

$$= 652.203 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	4186.19	Kgf
Transverse Shear Load	508.02	Kgf
Longitudinal Shear Load	354.13	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 460.000 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -4136 \cdot 1220.0 [1 - (1 - 1220.0/6226.8 + (252.587^2 - 0.0^2)/(2 \cdot 1220.0 \cdot 6226.8))/(1 + (4 \cdot 0.0)/(3 \cdot 6226.8))]$$

$$= -967.6 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 4136 \cdot 6227/4((1 + 2(253^2 - 0^2)/(6227^2))/(1 + (4 \cdot 0)/(3 \cdot 6227)) - 4 \cdot 1220/6227)$$

$$= 1414.0 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 17.261 \cdot 252.587/(2 \cdot 8.825) - 1414.0/(\pi \cdot 252.6^2 \cdot 8.825)$$

$$= 16.86 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$



$$= 17.261 \cdot 252.587 / (2 \cdot 8.825) + 1414.0 / (\pi \cdot 252.6^2 \cdot 8.825)$$

$$= 32.54 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 17.261 \cdot 252.587 / (2 \cdot 8.825) - 967.6 / (0.1066 \cdot \pi \cdot 252.6^2 \cdot 8.825)$$

$$= 75.02 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 17.261 \cdot 252.587 / (2 \cdot 8.825) + 967.6 / (0.1923 \cdot \pi \cdot 252.6^2 \cdot 8.825)$$

$$= -3.19 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 4136(6226.8 - 2 \cdot 1220.0) / (6226.8 + (4 \cdot 0.0 / 3))$$

$$= 2515.6 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 2515.57 / (252.5875 \cdot 8.825)$$

$$= 12.96 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(252.587 \cdot 8.825)}$$

$$= 36.826 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{(252.587 \cdot 8.825)}, 2 \cdot 1220.0)$$

$$= 223.65 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(137.997 / 137.997, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 4136 \cdot 0.1 / (223.653(8.825 + 1 \cdot 12.0))$$

$$= -0.66 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -4136 / (4(8.825 + 1 \cdot 12.0)223.653) - 3 \cdot 0.053 \cdot 4136 / (2(8.825 + 1 \cdot 12.0)^2)$$

$$= -9.59 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 3656.0 \text{ mm.}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000012 \cdot 3656.001(132.2 - 21.1)$$

$$= 5.001 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	460.0000	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	150.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	2	
Rib Thickness	Ribtk	12.0000	mm.
Web Thickness	Webtk	12.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	234.1	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.450	



Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(248.175 \cdot 8.825)} + 225.0)^3) / 12$$

$$= 1.707 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 1.707 + 26.299 (93.88 - 4.412)^2$$

$$= 2106.781 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	298.0	8.8	4.4	26.3	11604.5	0.211E+04
Wearplate	225.0	12.0	14.8	27.0	40027.5	0.169E+04
Web	12.0	215.0	128.3	25.8	331078.6	0.130E+04
BasePlate	150.0	16.0	243.8	24.0	585180.1	0.540E+04
Totals	...	...	...	103.1	967890.6	0.105E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 967891 / 103.099$$

$$= 93.880 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 4136.47$$

$$= 841.8613 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.0750 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 140.4785 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 500.0 - 248.17 \cdot \sin(120.0 / 2) / 1.0472$$

$$= 294.761 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 248.1527 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 2.1761 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 156.0872 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(3(4136 + 50) \cdot 150.0) / (2 \cdot 460.0 \cdot 156.087)}$$

$$= 11.342 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0 / 2) (245.0 + 12.0 + 12.0)$$

$$= 465.922 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 465.9216 / (2 - 1)$$

$$= 465.922 \text{ mm.}$$



Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 465.9216 \cdot 150.0 / 2$$

$$= 349.441 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 4186.192 / (460.0 \cdot 150.0)$$

$$= 6.067 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 349.4 \cdot 6.07$$

$$= 2120.041 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 16.56 + 27.955$$

$$= 44.515 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 2120.0 / 44.5153$$

$$= 4.671 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	12.0	138.0	81.0	16.6	134136.0	630.
Web	233.0	12.0	6.0	28.0	16773.2	221.
Totals	...	...	...	44.5	150909.2	851.

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 150.0 - 12.0$$

$$= 138.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 150909.172 / 44.515$$

$$= 33.901 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, Saddle \text{ Width} - ytot)$$

$$= \max(33.901, 150.0 - 33.901)$$

$$= 116.099 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (851.1 / 44.515) }$$

$$= 43.727 \text{ mm.}$$

Length of Outer Rib, step 1, [Lv]:

$$= Saddle \text{ Height} - \cos(saddle \text{ bearing angle} / 2)(radius + shlthk + wpdthk) - bpthk$$

$$= 500.0 - \cos(120.0 / 2)(245.0 + 12.0 + 12.0) - 16.0$$

$$= 349.500 \text{ mm.}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(saddle \text{ bearing angle} / 2)(radius + shlthk + wpdthk)$$

$$= 2 \cdot \sin(120.0 / 2)(245.0 + 12.0 + 12.0)$$

$$= 465.922 \text{ mm.}$$

Length of Outer Rib, step 3, [Bd]:

$$= Baseplate \text{ length} - clearance$$

$$= 460.0 - 50.8$$

$$= 409.200 \text{ mm.}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (465.922 - 409.2) / 2$$

$$= 28.361 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \sqrt{(Lv^2 + Dd^2)}$$

$$= \sqrt{(349.5^2 + 28.361^2)}$$

$$= 350.649 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19615E+09 / 234.1)}$$

$$= 128.601$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 350.649 / 43.727$$

$$= 8.019$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 354.1 / (2 \cdot 460.0) \cdot 465.922 \cdot 350.65 / 2$$

$$= 31.444 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($8.0191 < 128.6011$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (8.02)^2 / (2 \cdot 128.6^2)) 234 / (5/3 + 3 \cdot (8.02) / (8 \cdot 128.6) - (8.02^3) / (8 \cdot 128.6^3))$$

$$= 138.3 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc / Sca + (Rm \cdot C1 / I) / Sba$$

$$= 4.67 / 138.27 + (31.44 \cdot 116.099 / 8511374) / 156.09$$

$$= 0.061$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	137.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
Nominal Bolt Diameter	Bnd	25.4000 mm.
Thread Series	Series	TEMA
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BlArea	2.6238 cm ²
Saddle Load Q0 (Weight)	Q0	2529.6 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	48.4 Kg
Maximum Transverse Force	Ft	508.0 Kg
Maximum Longitudinal Force	Fl	354.1 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 354.1 / (2.624 \cdot 2)$$

$$= 6.6 \text{ N./mm}^2. \text{ Must be less than } 107.9 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 508.0 / (2.624 \cdot 2)$$

$$= 9.5 \text{ N./mm}^2. \text{ Must be less than } 107.9 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction)}$$



Bolt Area due to Shear Load [Bltarears]:
= $F_I / (BoltShearAllowable \cdot Nbolts)$
= $354.13 / (107.87 \cdot 2.0)$
= 0.1610 cm^2

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot Ft + \text{Sum of X Moments}$
= $500.0 \cdot 508.02 + 0.0$
= 254.02 Kg-m.

Eccentricity (e):
= $Rmom / QO$
= $254.02 / 2529.6$
= $100.42 \text{ mm.} > Bplen / 6 \rightarrow \text{Uplift in Transverse direction}$

$f = Bplen / 2 - Edgedis$
= $460.0 / 2 - 50.8$
= 179.20 mm.

Modular Ratio Of Steel/Concrete (n1):
= ES / EC
= $203402.5 / 21526.32$
= 9.45

$K1 = 3 (e - 0.5 \cdot Bplen)$
= $3 (100.42 - 0.5 \cdot 460.0)$
= -388.75 mm.

$K2 = 6 \cdot n1 \cdot At / Bpwid \cdot (f + e)$
= $6 \cdot 9.45 \cdot 2.62 / 150.0 \cdot (179.2 + 100.42)$
= 27728.98 mm.^2

$K3 = -K2 \cdot (0.5 \cdot Bplen + f)$
= $-27728.97 \cdot (0.5 \cdot 460.0 + 179.2)$
= $-11346696.78 \text{ mm.}^3$

Iteratively Solving for the Effective Bearing Length:
 $Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$
 $Y^3 + -388.75 \cdot Y^2 + 27728.97 \cdot Y + -0.1E+08 = 0$
 $Y = 391.88 \text{ mm.}$

$Num = (Bplen / 2 - Y / 3 - e)$
= $(460.0 / 2 - 391.88 / 3 - 100.42)$
= -1.04

$Denom = (Bplen / 2 - Y / 3 + f)$
= $(460.0 / 2 - 391.88 / 3 + 179.2)$
= 278.57

Total Bolt Tension Force [Tforce]:
= $-QO \cdot Num / Denom$
= $-2529.6 \cdot -1.04 / 278.57$
= 9.47 Kg

Bolt Area Required due to Transverse Load [Bltareart]:
= $Tforce / (Stba \cdot Nbt)$
= $9.47 / (137.3 \cdot 1.0)$
= 0.0068 cm^2

Required Area of a Single Bolt [Bltarear]:
= $\max[Bltarearl, Bltarears, Bltareart]$
= $\max[0.0, 0.161, 0.0068]$
= 0.1610 cm^2

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)
 $= 2(QO + Tforce) / (Y \cdot Bpwid)$
 $= 2(2529.6 + 9.47) / (391.88 \cdot 150.0)$
 $= 8.47 \text{ bars}$

Distance from Baseplate Edge to the Web [ADIST]:
 $= (Bplen - Weblngh) / 2$
 $= (460.0 - 409.2) / 2$
 $= 25.4000 \text{ mm.}$

Overturning Moment due To Bolt Tension [Mt]:
 $= Tforce \cdot Adist$
 $= 9.47 \cdot 25.4$
 $= 0.24 \text{ Kg-m.}$

Equivalent Bearing Pressure (f1):
 $= fc \cdot (Y - Adist) / Y$
 $= 8.47 \cdot (391.88 - 25.4) / 391.88$
 $= 7.92 \text{ bars}$

Overturning Moment due to Bearing Pressure [Mc]:
 $= (Adist^2 \cdot Bpwid / 6) \cdot (f1 + 2 \cdot fc)$
 $= (25.4^2 \cdot 150.0 / 6) \cdot (7.92 + 2 \cdot 8.47)$
 $= 4.09 \text{ Kg-m.}$

Baseplate Required Thickness [Treq]:
 $= \sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))}$
 $= \sqrt{(6 \cdot \max(0.24, 4.09) / (150.0 \cdot 207.0))}$
 $= 2.7840 \text{ mm.}$

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 12.0, tr = 5.092, c = 3.175 mm., E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.577, Temp. Reduction = 24 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -47 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	252.59	mm.
Shell Thickness used in this Case	t	12.000	mm.
Stiffened Vessel Length per 4.15.6	L	6226.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	1220.00	mm.
Saddle Width	b1 or b	150.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	225.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	130.09	mm.
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	500.00	mm.
Coefficient of Friction	mu	0.45	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case 3185.92 Kgf



Pressure used in Analysis P 17.261 bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	19.44	138.00
Long. Stress at Bottom of Midspan	29.97	138.00
Long. Stress at Top of Saddles	68.10	138.00
Long. Stress at Bottom of Saddles	0.65	138.00
Tangential Shear in Shell	9.71	110.40
Circ. Stress at Horn of Saddle	7.39	172.50
Circ. Compressive Stress in Shell	0.51	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(1016.0/2 + 0) \cdot 500.0/460.0$
 $= 1656.6\ Kg$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(66.51, 0) \cdot 500.0/3656.0007$
 $= 9.1\ Kg$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(354.13, 0) \cdot 500.0/3656.0007$
 $= 48.4\ Kg$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(354/2 + 0) \cdot 500.0/460.0$
 $= 577.4\ Kg$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 1529 + \max(9, 1657, 48, 577)$
 $= 3185.9\ Kg$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 1984.306(0.0 + \pi/4(0.514 \cdot 1.2)^2)$
 $= 652.203\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	3235.65	Kgf
Transverse Shear Load	508.02	Kgf
Longitudinal Shear Load	354.13	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= Baseplate\ Length$
 $= 460.000\ mm.$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$



$$= -3186 \cdot 1220.0 [1 - (1 - 1220.0/6226.8 + (252.587^2 - 130.087^2) / (2 \cdot 1220.0 \cdot 6226.8)) / (1 + (4 \cdot 130.09) / (3 \cdot 6226.8))] \\ = -834.6 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a/L) \\ = 3186 \cdot 6227 / 4 \cdot ((1 + 2(253^2 - 130^2) / (6227^2)) / (1 + (4 \cdot 130) / (3 \cdot 6227)) - 4 \cdot 1220 / 6227) \\ = 950.0 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t) \\ = 17.261 \cdot 252.587 / (2 \cdot 8.825) - 950.0 / (\pi \cdot 252.6^2 \cdot 8.825) \\ = 19.44 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\ = 17.261 \cdot 252.587 / (2 \cdot 8.825) + 950.0 / (\pi \cdot 252.6^2 \cdot 8.825) \\ = 29.97 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\ = 17.261 \cdot 252.587 / (2 \cdot 8.825) - 834.6 / (0.1066 \cdot \pi \cdot 252.6^2 \cdot 8.825) \\ = 68.10 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\ = 17.261 \cdot 252.587 / (2 \cdot 8.825) + 834.6 / (0.1923 \cdot \pi \cdot 252.6^2 \cdot 8.825) \\ = 0.65 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\ = 3186(6226.8 - 2 \cdot 1220.0) / (6226.8 + (4 \cdot 130.09 / 3)) \\ = 1885.0 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t) \\ = 1.1707 \cdot 1885.0 / (252.587 \cdot 8.825) \\ = 9.71 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)} \\ = 0.78 \cdot \sqrt{(252.587 \cdot 8.825)} \\ = 36.826 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\ = \min(150.0 + 1.56 \cdot \sqrt{(252.587 \cdot 8.825)}, 2 \cdot 1220.0) \\ = 223.65 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(Sr/S, 1) \\ = \min(137.997/137.997, 1) \\ = 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\ = -0.7603 \cdot 3186 \cdot 0.1 / (223.653(8.825 + 1 \cdot 12.0)) \\ = -0.51 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2) \\ = -3186 / (4(8.825 + 1 \cdot 12.0)223.653) - 3 \cdot 0.053 \cdot 3186 / (2(8.825 + 1 \cdot 12.0)^2) \\ = -7.39 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	460.0000	mm.
Baseplate Thickness	Bpthk	16.0000	mm.

Baseplate Width	Bpwid	150.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	2	
Rib Thickness	Ribtk	12.0000	mm.
Web Thickness	Webtk	12.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	234.1	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.450	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}}) t^3) / 12$
 $= ((1.56 \cdot \sqrt{(248.175 \cdot 8.825) + 225.0}) 8.825^3) / 12$
 $= 1.707 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 1.707 + 26.299 (93.88 - 4.412)^2$
 $= 2106.781 \text{ cm}^4$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	298.0	8.8	4.4	26.3	11604.5	0.211E+04
Wearplate	225.0	12.0	14.8	27.0	40027.5	0.169E+04
Web	12.0	215.0	128.3	25.8	331078.6	0.130E+04
BasePlate	150.0	16.0	243.8	24.0	585180.1	0.540E+04
Totals	...	...	...	103.1	967890.6	0.105E+05

Distance to Centroid [C1]:
 $= AY / A$
 $= 967891 / 103.099$
 $= 93.880 \text{ mm.}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 120.0 / 2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.204 \cdot 3185.925$
 $= 648.4047 \text{ Kg}$

Tension Stress, St = (Fh/As) = 0.8280 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 140.4785 N./mm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 500.0 - 248.17 \cdot \sin(120.0/2) / 1.0472$
 $= 294.761 \text{ mm.}$

Bending Moment, M = Fh * d = 191.1282 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 1.6760 N./mm²
 Allowed Stress, Sa = 2/3 * Yield Str = 156.0872 N./mm²

Minimum Thickness of Baseplate per Moss:

$= \sqrt{ (3 (Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})) }$
 $= \sqrt{ (3 (3186 + 50) 150.0 / (2 \cdot 460.0 \cdot 156.087)) }$
 $= 9.972 \text{ mm.}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (245.0 + 12.0 + 12.0)$$

$$= 465.922 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 465.9216 / (2 - 1)$$

$$= 465.922 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 465.9216 \cdot 150.0 / 2$$

$$= 349.441 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 3235.648 / (460.0 \cdot 150.0)$$

$$= 4.689 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 349.4 \cdot 4.69$$

$$= 1638.650 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 16.56 + 27.955$$

$$= 44.515 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 1638.7 / 44.5153$$

$$= 3.610 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	12.0	138.0	81.0	16.6	134136.0	630.
Web	233.0	12.0	6.0	28.0	16773.2	221.
Totals	...	...	...	44.5	150909.2	851.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 150.0 - 12.0$$

$$= 138.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 150909.172 / 44.515$$

$$= 33.901 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(33.901, 150.0 - 33.901)$$

$$= 116.099 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(851.1 / 44.515)}$$

$$= 43.727 \text{ mm.}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 500.0 - \cos(120.0/2) (245.0 + 12.0 + 12.0) - 16.0$$

$$= 349.500 \text{ mm.}$$



Length of Outer Rib, step 2, [Lw]:
 $= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$
 $= 2 \cdot \sin(120.0/2) (245.0 + 12.0 + 12.0)$
 $= 465.922 \text{ mm.}$

Length of Outer Rib, step 3, [Bd]:
 $= \text{Baseplate length} - \text{clearance}$
 $= 460.0 - 50.8$
 $= 409.200 \text{ mm.}$

Length of Outer Rib, step 4, [Dd]:
 $= (Lw - Bd) / 2$
 $= (465.922 - 409.2) / 2$
 $= 28.361 \text{ mm.}$

Length of Outer Rib [L]:
 $= \sqrt{Lv^2 + Dd^2}$
 $= \sqrt{349.5^2 + 28.361^2}$
 $= 350.649 \text{ mm.}$

Intermediate Term [Cc]:
 $= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 $= \sqrt{(2 \cdot \pi^2 \cdot 0.19615E+09 / 234.1)}$
 $= 128.601$

Slenderness ratio [KL/r]:
 $= KL/r$
 $= 1 \cdot 350.649 / 43.727$
 $= 8.019$

Bending Moment [Rm]:
 $= Fl / (2 \cdot Bplen) \cdot e \cdot L / 2$
 $= 354.1 / (2 \cdot 460.0) \cdot 465.922 \cdot 350.65 / 2$
 $= 31.444 \text{ Kg-m.}$

Compressive Allowable, $KL/r < Cc$ ($8.0191 < 128.6011$) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (8.02)^2 / (2 \cdot 128.6^2)) 234 /$
 $(5/3 + 3 \cdot (8.02) / (8 \cdot 128.6) - (8.02^3) / (8 \cdot 128.6^3))$
 $= 138.3 \text{ N./mm}^2$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$= Sc/Sca + (Rm \cdot C1 / I) / Sba$
 $= 3.61 / 138.27 + (31.44 \cdot 116.099 / 8511374) / 156.09$
 $= 0.053$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	137.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
Nominal Bolt Diameter	Bnd	25.4000 mm.
Thread Series	Series	TEMA
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BlArea	2.6238 cm ²
Saddle Load Q0 (Weight)	Q0	1579.1 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	48.4 Kg
Maximum Transverse Force	Ft	508.0 Kg
Maximum Longitudinal Force	Fl	354.1 Kg
Saddle Bolted to Steel Foundation	No	

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
 $= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 354.1 / (2.624 \cdot 2)$
 $= 6.6 \text{ N./mm}^2$. Must be less than 107.9 N./mm².



Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $508.0 / (2.624 \cdot 2)$
= 9.5 N./mm^2 . Must be less than 107.9 N./mm^2 .

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $354.13 / (107.87 \cdot 2.0)$
= 0.1610 cm^2

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $500.0 \cdot 508.02 + 0.0$
= 254.02 Kg-m .

Eccentricity (e):
= R_{mom} / QO
= $254.02 / 1579.05$
= 160.86 mm . > $B_{plen} / 6$ --> Uplift in Transverse direction

$f = B_{plen} / 2 - \text{Edgedis}$
= $460.0 / 2 - 50.8$
= 179.20 mm .

Modular Ratio Of Steel/Concrete (n1):
= E_S / E_C
= $203402.5 / 21526.32$
= 9.45

$K1 = 3 (e - 0.5 \cdot B_{plen})$
= $3 (160.86 - 0.5 \cdot 460.0)$
= -207.41 mm .

$K2 = 6 \cdot n1 \cdot A_t / B_{pwid} \cdot (f + e)$
= $6 \cdot 9.45 \cdot 2.62 / 150.0 \cdot (179.2 + 160.86)$
= 33723.42 mm^2 .

$K3 = -K2 \cdot (0.5 \cdot B_{plen} + f)$
= $-33723.42 \cdot (0.5 \cdot 460.0 + 179.2)$
= $-13799622.64 \text{ mm}^3$.

Iteratively Solving for the Effective Bearing Length:
 $Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$
 $Y^3 + -207.41 \cdot Y^2 + 33723.42 \cdot Y + -0.1E+08 = 0$
 $Y = 270.93 \text{ mm}$.

$Num = (B_{plen} / 2 - Y / 3 - e)$
= $(460.0 / 2 - 270.93 / 3 - 160.86)$
= -21.17

$Denom = (B_{plen} / 2 - Y / 3 + f)$
= $(460.0 / 2 - 270.93 / 3 + 179.2)$
= 318.89

Total Bolt Tension Force [Tforce]:
= $-QO \cdot Num / Denom$
= $-1579.05 \cdot -21.17 / 318.89$
= 104.85 Kg

Bolt Area Required due to Transverse Load [Bltareart]:
= $T_{force} / (Stba \cdot Nbt)$
= $104.85 / (137.3 \cdot 1.0)$



$$= 0.0749 \text{ cm}^2$$

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0.0, 0.161, 0.0749]
= 0.1610 cm²

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)
= $2(QO + Tforce) / (Y \cdot Bpwid)$
= $2(1579.05 + 104.85) / (270.93 \cdot 150.0)$
= 8.13 bars

Distance from Baseplate Edge to the Web [ADIST]:
= $(Bplen - WebIngh) / 2$
= $(460.0 - 409.2) / 2$
= 25.4000 mm.

Overturning Moment due To Bolt Tension [Mt]:
= $Tforce \cdot Adist$
= $104.85 \cdot 25.4$
= 2.66 Kg-m.

Equivalent Bearing Pressure (f1):
= $fc \cdot (Y - Adist) / Y$
= $8.13 \cdot (270.93 - 25.4) / 270.93$
= 7.36 bars

Overturning Moment due to Bearing Pressure [Mc]:
= $(Adist^2 \cdot Bpwid / 6) \cdot (f1 + 2 \cdot fc)$
= $(25.4^2 \cdot 150.0 / 6) \cdot (7.36 + 2 \cdot 8.13)$
= 3.88 Kg-m.

Baseplate Required Thickness [Treq]:
= $\sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))}$
= $\sqrt{(6 \cdot \max(2.66, 3.88) / (150.0 \cdot 207.0))}$
= 2.7132 mm.

MDMT Calculations for: RIGHT SADDLE, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 12.0, tr = 5.092, c = 3.175 mm., E* = 1.0
Thickness Ratio = $tr \cdot E* / (tg - c) = 0.577$, Temp. Reduction = 24 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-47 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Review notes about nozzle loadings and supports in the Warnings report.

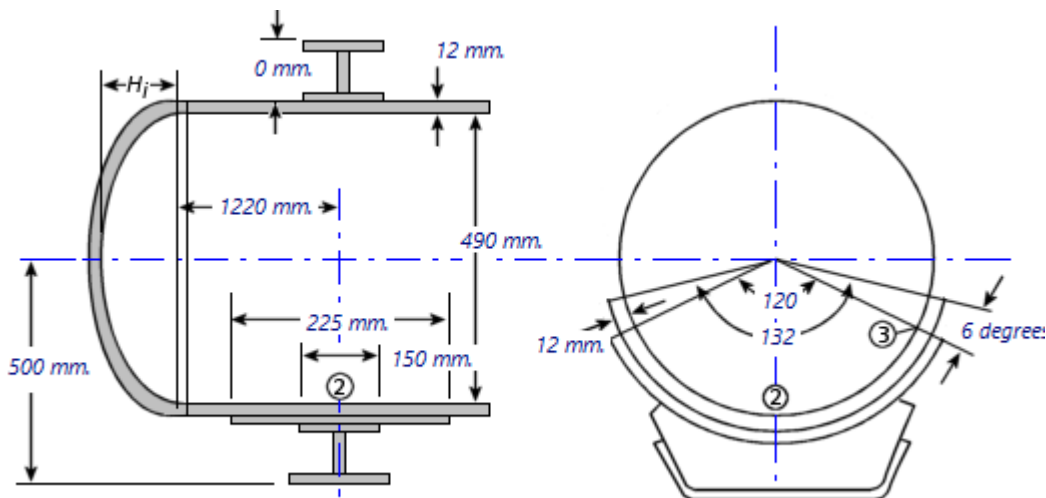
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	252.59	mm.
Shell Thickness used in this Case	t	12.000	mm.
Stiffened Vessel Length per 4.15.6	L	6226.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	1220.00	mm.
Saddle Width	b1 or b	150.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	225.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		247.00	N./mm ²
Head Allowable Stress used in Calculation		247.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	500.00	mm.
Coefficient of Friction	mu	0.45	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		3082.58	Kgf
Pressure used in Analysis	P	36.385	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	46.23	247.00
Long. Stress at Bottom of Midspan	57.92	247.00
Long. Stress at Top of Saddles	89.57	247.00
Long. Stress at Bottom of Saddles	31.29	247.00
Tangential Shear in Shell	9.66	138.58
Circ. Stress at Horn of Saddle	12.11	370.49
Circ. Compressive Stress in Shell	0.66	247.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft}/\text{Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(335.3/2 + 0) \cdot 500.0/460.0$$

$$= 546.7 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(21.95, 0) \cdot 500.0/3656.0007$$

$$= 3.0 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 2536 + \max(3, 547, 0, 0)$$

$$= 3082.6 \text{ Kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1984.306(0.0 + \pi/4(0.514 \cdot 1.2)^2)$$

$$= 652.203 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	3132.30	Kgf
Transverse Shear Load	167.65	Kgf
Longitudinal Shear Load	21.95	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 460.000 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -3083 \cdot 1220.0 [1 - (1 - 1220.0/6226.8 + (252.587^2 - 0.0^2)/(2 \cdot 1220.0 \cdot 6226.8))/(1 + (4 \cdot 0.0)/(3 \cdot 6226.8))]$$

$$= -721.1 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 3083 \cdot 6227/4((1 + 2(253^2 - 0^2)/(6227^2))/(1 + (4 \cdot 0)/(3 \cdot 6227)) - 4 \cdot 1220/6227)$$

$$= 1053.7 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 36.385 \cdot 252.587/(2 \cdot 8.825) - 1053.7/(\pi \cdot 252.6^2 \cdot 8.825)$$

$$= 46.23 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 36.385 \cdot 252.587/(2 \cdot 8.825) + 1053.7/(\pi \cdot 252.6^2 \cdot 8.825)$$

$$= 57.92 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 36.385 \cdot 252.587/(2 \cdot 8.825) - 721.1/(0.1066 \cdot \pi \cdot 252.6^2 \cdot 8.825)$$

$$= 89.57 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 36.385 \cdot 252.587/(2 \cdot 8.825) + 721.1/(0.1923 \cdot \pi \cdot 252.6^2 \cdot 8.825)$$



= 31.29 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:
 = $Q(L-2a)/(L+(4 \cdot hm/3))$
 = $3083(6226.8 - 2 \cdot 1220.0)/(6226.8 + (4 \cdot 0.0/3))$
 = 1874.7 Kgf

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:
 = $K_2 \cdot T / (R_m \cdot t)$
 = $1.1707 \cdot 1874.66 / (252.5875 \cdot 8.825)$
 = 9.66 N./mm²

Decay Length (4.15.20) [x1,x2]:
 = $0.78 \cdot \sqrt{R_m \cdot t}$
 = $0.78 \cdot \sqrt{252.587 \cdot 8.825}$
 = 36.826 mm.

Effective reinforcing plate width (4.15.24) [B1]:
 = $\min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 = $\min(150.0 + 1.56 \cdot \sqrt{252.587 \cdot 8.825}, 2 \cdot 1220.0)$
 = 223.65 mm.

Wear Plate/Shell Stress ratio (4.15.26) [eta]:
 = $\min(S_r/S, 1)$
 = $\min(137.997/246.996, 1)$
 = 0.5587

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:
 = $-K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$
 = $-0.7603 \cdot 3083 \cdot 0.1 / (223.653(8.825 + 0.559 \cdot 12.0))$
 = -0.66 N./mm²

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:
 = $-Q/(4(t+\eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q/(2(t+\eta \cdot tr)^2)$
 = $-3083/(4(8.825 + 0.559 \cdot 12.0)223.653) - 3 \cdot 0.053 \cdot 3083/(2(8.825 + 0.559 \cdot 12.0)^2)$
 = -12.11 N./mm²

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	460.0000	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	150.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	2	
Rib Thickness	Ribtk	12.0000	mm.
Web Thickness	Webtk	12.0000	mm.
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.450	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 = $((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth})t^3)/12$
 = $((1.56 \cdot \sqrt{248.175 \cdot 8.825}) + 225.0)8.825^3/12$
 = 1.707 cm⁴

Resolved Inertia for the Shell [I,shell]:
 = $\text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$
 = $1.707 + 26.299(93.88 - 4.412)^2$
 = 2106.781 cm⁴

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	298.0	8.8	4.4	26.3	11604.5	0.211E+04
Wearplate	225.0	12.0	14.8	27.0	40027.5	0.169E+04
Web	12.0	215.0	128.3	25.8	331078.6	0.130E+04
BasePlate	150.0	16.0	243.8	24.0	585180.1	0.540E+04
Totals	...	...	...	103.1	967890.6	0.105E+05

Distance to Centroid [C1]:

$$\begin{aligned} &= AY / A \\ &= 967891/103.099 \\ &= 93.880 \text{ mm.} \end{aligned}$$

Angle [β]:

$$\begin{aligned} &= 180 - \text{Saddle Angle}/2 \\ &= 180 - 120.0/2 \\ &= 120.0 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta)) \\ &= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0)) \\ &= 0.2035 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.204 \cdot 3082.577 \\ &= 627.3712 \text{ Kg} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh/As) = 0.8011 \text{ N./mm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\ &= 500.0 - 248.17 \cdot \sin(120.0/2) / 1.0472 \\ &= 294.761 \text{ mm.} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 184.9282 \text{ Kg-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 1.6217 \text{ N./mm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\ &= \sqrt{3(3083 + 50)150.0 / (2 \cdot 460.0 \cdot 173.33)} \\ &= 9.310 \text{ mm.} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120.0/2) (245.0 + 12.0 + 12.0) \\ &= 465.922 \text{ mm.} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 465.9216 / (2 - 1) \\ &= 465.922 \text{ mm.} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 465.9216 \cdot 150.0/2 \\ &= 349.441 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 3132.3 / (460.0 \cdot 150.0) \\ &= 4.540 \text{ Kg/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 349.4 \cdot 4.54 \\ &= 1586.311 \text{ Kg} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 16.56 + 27.955 \\ &= 44.515 \text{ cm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 1586.3/44.5153$$

$$= 3.495 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	12.0	138.0	81.0	16.6	134136.0	630.
Web	233.0	12.0	6.0	28.0	16773.2	221.
Totals	...	...	...	44.5	150909.2	851.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 150.0 - 12.0$$

$$= 138.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 150909.172/44.515$$

$$= 33.901 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{Saddle Width} - y_{tot})$$

$$= \max(33.901, 150.0 - 33.901)$$

$$= 116.099 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(851.1/44.515)}$$

$$= 43.727 \text{ mm.}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 500.0 - \cos(120.0/2)(245.0 + 12.0 + 12.0) - 16.0$$

$$= 349.500 \text{ mm.}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120.0/2)(245.0 + 12.0 + 12.0)$$

$$= 465.922 \text{ mm.}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 460.0 - 50.8$$

$$= 409.200 \text{ mm.}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd)/2$$

$$= (465.922 - 409.2)/2$$

$$= 28.361 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \sqrt{(L_v^2 + D_d^2)}$$

$$= \sqrt{(349.5^2 + 28.361^2)}$$

$$= 350.649 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19615\text{E}+09/260.0)}$$

$$= 122.037$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 350.649/43.727$$

$$= 8.019$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bpln) \cdot e \cdot L / 2$$

$$= 21.9 / (2 \cdot 460.0) \cdot 465.922 \cdot 350.65 / 2$$

$$= 1.949 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($8.0191 < 122.0369$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (8.02)^2 / (2 \cdot 122.04^2)) 260 /$$

$$(5/3 + 3 \cdot (8.02) / (8 \cdot 122.04) - (8.02^3) / (8 \cdot 122.04^3))$$

$$= 153.4 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 3.49 / 153.4 + (1.95 \cdot 116.099 / 8511374) / 173.33$$

$$= 0.024$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
Nominal Bolt Diameter	Bnd	25.4000 mm.
Thread Series	Series	TEMA
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	2.6238 cm ²
Saddle Load Q0 (Weight)	Q0	2585.6 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	3.0 Kg
Maximum Transverse Force	Ft	167.6 Kg
Maximum Longitudinal Force	F1	21.9 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= FI / (Bolt Area \cdot Number of Bolts)$$

$$= 21.9 / (2.624 \cdot 2)$$

$$= 0.4 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (Bolt Area \cdot Number of Bolts)$$

$$= 167.6 / (2.624 \cdot 2)$$

$$= 3.1 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (Q0 > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= FI / (BoltShearAllowable \cdot Nbolts)$$

$$= 21.95 / (103.2 \cdot 2.0)$$

$$= 0.0104 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 500.0 \cdot 167.65 + 0.0$$

$$= 83.83 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 83.83 / 2585.62$$

$$= 32.42 \text{ mm.} < Bpln / 6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0.0, 0.0104, 0.0]
 = 0.0104 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	252.59	mm.
Shell Thickness used in this Case	t	12.000	mm.
Stiffened Vessel Length per 4.15.6	L	6226.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	1220.00	mm.
Saddle Width	b1 or b	150.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	225.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	130.09	mm.
Shell Allowable Stress used in Calculation		247.00	N./mm ²
Head Allowable Stress used in Calculation		247.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	500.00	mm.
Coefficient of Friction	mu	0.45	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		2085.28	Kgf
Pressure used in Analysis	P	36.385	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	48.63	247.00
Long. Stress at Bottom of Midspan	55.52	247.00
Long. Stress at Top of Saddles	80.48	247.00
Long. Stress at Bottom of Saddles	36.33	247.00
Tangential Shear in Shell	6.35	138.58
Circ. Stress at Horn of Saddle	8.19	370.49
Circ. Compressive Stress in Shell	0.45	247.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 = Ftr(Ft/Num of Saddles + Z Force Load) • B / E
 = 3.0(335.3/2 + 0) • 500.0/460.0
 = 546.7 Kgf

Longitudinal Saddle Reaction Force [Fwl]:
 = max(Fl,wind; Sum of X Forces) • B / Ls
 = max(21.95, 0) • 500.0/3656.0007
 = 3.0 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 1539 + max(3, 547, 0, 0)
 = 2085.3 Kgf

Longitudinal Wind Force [Fl]:
 = WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))



$$= 1.1 \cdot 1984.306(0.0 + \pi/4(0.514 \cdot 1.2)^2)$$

$$= 652.203 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	2135.01	Kgf
Transverse Shear Load	167.65	Kgf
Longitudinal Shear Load	21.95	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= Baseplate Length
 = 460.000 mm.

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -2085 \cdot 1220.0 [1 - (1 - 1220.0/6226.8 + (252.587^2 - 130.087^2)/(2 \cdot 1220.0 \cdot 6226.8))/(1 + (4 \cdot 130.09)/(3 \cdot 6226.8))]$$

$$= -546.3 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 2085 \cdot 6227/4 ((1 + 2(253^2 - 130^2)/(6227^2))/(1 + (4 \cdot 130)/(3 \cdot 6227)) - 4 \cdot 1220/6227)$$

$$= 621.8 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 36.385 \cdot 252.587/(2 \cdot 8.825) - 621.8/(\pi \cdot 252.6^2 \cdot 8.825)$$

$$= 48.63 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 36.385 \cdot 252.587/(2 \cdot 8.825) + 621.8/(\pi \cdot 252.6^2 \cdot 8.825)$$

$$= 55.52 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 36.385 \cdot 252.587/(2 \cdot 8.825) - 546.3/(0.1066 \cdot \pi \cdot 252.6^2 \cdot 8.825)$$

$$= 80.48 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m/(2t) + M1/(K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 36.385 \cdot 252.587/(2 \cdot 8.825) + 546.3/(0.1923 \cdot \pi \cdot 252.6^2 \cdot 8.825)$$

$$= 36.33 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 2085(6226.8 - 2 \cdot 1220.0)/(6226.8 + (4 \cdot 130.09/3))$$

$$= 1233.8 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 1233.79 / (252.5875 \cdot 8.825)$$

$$= 6.35 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(252.587 \cdot 8.825)}$$

$$= 36.826 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:
 $= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$
 $= \min(150.0 + 1.56 \cdot \sqrt{ 252.587 \cdot 8.825 }, 2 \cdot 1220.0)$
 $= 223.65 \text{ mm.}$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:
 $= \min(S_r/S, 1)$
 $= \min(137.997/246.996, 1)$
 $= 0.5587$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:
 $= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$
 $= -0.7603 \cdot 2085 \cdot 0.1 / (223.653(8.825 + 0.559 \cdot 12.0))$
 $= -0.45 \text{ N./mm}^2$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:
 $= -Q / (4(t + \eta \cdot tr) b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$
 $= -2085 / (4(8.825 + 0.559 \cdot 12.0) 223.653) -$
 $3 \cdot 0.053 \cdot 2085 / (2(8.825 + 0.559 \cdot 12.0)^2)$
 $= -8.19 \text{ N./mm}^2$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	460.0000	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	150.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	2	
Rib Thickness	Ribtk	12.0000	mm.
Web Thickness	Webtk	12.0000	mm.
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.450	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{ R \cdot t } + \text{WearPlateWidth}) t^3) / 12$
 $= ((1.56 \cdot \sqrt{ 248.175 \cdot 8.825 } + 225.0) 8.825^3) / 12$
 $= 1.707 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$
 $= 1.707 + 26.299(93.88 - 4.412)^2$
 $= 2106.781 \text{ cm}^4$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	298.0	8.8	4.4	26.3	11604.5	0.211E+04
Wearplate	225.0	12.0	14.8	27.0	40027.5	0.169E+04
Web	12.0	215.0	128.3	25.8	331078.6	0.130E+04
BasePlate	150.0	16.0	243.8	24.0	585180.1	0.540E+04
Totals	...	...	...	103.1	967890.6	0.105E+05

Distance to Centroid [C1]:
 $= AY / A$
 $= 967891/103.099$
 $= 93.880 \text{ mm.}$

Angle [β]:
 $= 180 - \text{Saddle Angle}/2$
 $= 180 - 120.0/2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$
 $= 0.2035$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 2085.283$$

$$= 424.4002 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 0.5419 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 500.0 - 248.17 \cdot \sin(120.0/2) / 1.0472$$

$$= 294.761 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 125.0991 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 1.0970 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(2085 + 50) \cdot 150.0 / (2 \cdot 460.0 \cdot 173.33)}$$

$$= 7.687 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) \cdot (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) \cdot (245.0 + 12.0 + 12.0)$$

$$= 465.922 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 465.9216 / (2 - 1)$$

$$= 465.922 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 465.9216 \cdot 150.0 / 2$$

$$= 349.441 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 2135.006 / (460.0 \cdot 150.0)$$

$$= 3.094 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 349.4 \cdot 3.09$$

$$= 1081.245 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 16.56 + 27.955$$

$$= 44.515 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 1081.2 / 44.5153$$

$$= 2.382 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	12.0	138.0	81.0	16.6	134136.0	630.
Web	233.0	12.0	6.0	28.0	16773.2	221.
Totals	...	...	...	44.5	150909.2	851.

Rib dimension [D]:

$$\begin{aligned} &= \text{Saddle Width} - \text{Web Thickness} \\ &= 150.0 - 12.0 \\ &= 138.000 \text{ mm.} \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned} &= AY / A \\ &= 150909.172 / 44.515 \\ &= 33.901 \text{ mm.} \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned} &= \max(y_{tot}, \text{Saddle Width} - y_{tot}) \\ &= \max(33.901, 150.0 - 33.901) \\ &= 116.099 \text{ mm.} \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned} &= \sqrt{ (\text{Total Inertia} / \text{Total Area}) } \\ &= \sqrt{ (851.1 / 44.515) } \\ &= 43.727 \text{ mm.} \end{aligned}$$

Length of Outer Rib, step 1, [Lv]:

$$\begin{aligned} &= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\ &= 500.0 - \cos(120.0 / 2) (245.0 + 12.0 + 12.0) - 16.0 \\ &= 349.500 \text{ mm.} \end{aligned}$$

Length of Outer Rib, step 2, [Lw]:

$$\begin{aligned} &= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) \\ &= 2 \cdot \sin(120.0 / 2) (245.0 + 12.0 + 12.0) \\ &= 465.922 \text{ mm.} \end{aligned}$$

Length of Outer Rib, step 3, [Bd]:

$$\begin{aligned} &= \text{Baseplate length} - \text{clearance} \\ &= 460.0 - 50.8 \\ &= 409.200 \text{ mm.} \end{aligned}$$

Length of Outer Rib, step 4, [Dd]:

$$\begin{aligned} &= (Lw - Bd) / 2 \\ &= (465.922 - 409.2) / 2 \\ &= 28.361 \text{ mm.} \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned} &= \sqrt{ (Lv^2 + Dd^2) } \\ &= \sqrt{ (349.5^2 + 28.361^2) } \\ &= 350.649 \text{ mm.} \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned} &= \sqrt{ (2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}) } \\ &= \sqrt{ (2 \cdot \pi^2 \cdot 0.19615\text{E}+09 / 260.0) } \\ &= 122.037 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned} &= KL / r \\ &= 1 \cdot 350.649 / 43.727 \\ &= 8.019 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned} &= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2 \\ &= 21.9 / (2 \cdot 460.0) \cdot 465.922 \cdot 350.65 / 2 \\ &= 1.949 \text{ Kg-m.} \end{aligned}$$

Compressive Allowable, $KL/r < Cc$ ($8.0191 < 122.0369$) per AISC E2-1 [Sca]:

$$\begin{aligned} &= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3)) \\ &= (1 - (8.02)^2 / (2 \cdot 122.04^2)) 260 / \\ &\quad (5/3 + 3 \cdot (8.02) / (8 \cdot 122.04) - (8.02^3) / (8 \cdot 122.04^3)) \\ &= 153.4 \text{ N./mm}^2 \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$\begin{aligned} &= Sc / Sca + (Rm \cdot C1 / I) / Sba \\ &= 2.38 / 153.4 + (1.95 \cdot 116.099 / 8511374) / 173.33 \end{aligned}$$



= 0.017

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
Nominal Bolt Diameter	Bnd	25.4000 mm.
Thread Series	Series	TEMA
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	2.6238 cm ²
Saddle Load Q0 (Weight)	Q0	1588.3 Kg _f
Saddle Load QL (Wind/Seismic contribution)	QL	3.0 Kg _f
Maximum Transverse Force	Ft	167.6 Kg _f
Maximum Longitudinal Force	Fl	21.9 Kg _f
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$\begin{aligned} &= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\ &= 21.9 / (2.624 \cdot 2) \\ &= 0.4 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2. \end{aligned}$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$\begin{aligned} &= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\ &= 167.6 / (2.624 \cdot 2) \\ &= 3.1 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2. \end{aligned}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$\begin{aligned} &= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts}) \\ &= 21.95 / (103.2 \cdot 2.0) \\ &= 0.0104 \text{ cm}^2 \end{aligned}$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$\begin{aligned} &= B \cdot Ft + \text{Sum of X Moments} \\ &= 500.0 \cdot 167.65 + 0.0 \\ &= 83.83 \text{ Kg-m.} \end{aligned}$$

Eccentricity (e):

$$\begin{aligned} &= Rmom / Q0 \\ &= 83.83 / 1588.33 \\ &= 52.77 \text{ mm.} < Bplen/6 \text{ --> No Uplift in Transverse direction} \end{aligned}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$\begin{aligned} &= \max[Bltarearl, Bltarears, Bltareart] \\ &= \max[0.0, 0.0104, 0.0] \\ &= 0.0104 \text{ cm}^2 \end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: TS IN NOZ T1

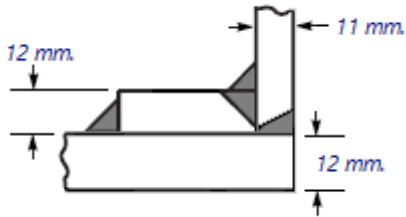
From: 20

Pressure for Reinforcement Calculations	P	5.564	bars
Temperature for Internal Pressure	Temp	77	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	490.00	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	c	3.1750	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		250.00	mm.
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.1750	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	268.2750	mm.
Thickness of Pad	te	12.0000	mm.
Weld leg size between Pad and Shell	Wp	10.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm.
Reinforcing Pad Width		50.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: TS IN NOZ T1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	5.761 in.
Actual Thickness Used in Calculation	0.432 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.56 \cdot 248.175 / (138 \cdot 1.0 - 0.6 \cdot 5.56)$
 $= 1.0031$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.56 \cdot 76.34 / (118 \cdot 1.0 - 0.6 \cdot 5.56)$
 $= 0.3610$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	305.3588 mm.
Parallel to Vessel Wall, opening length	d	152.6794 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		22.0625 mm.

Weld Strength Reduction Factor [fr1]:
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

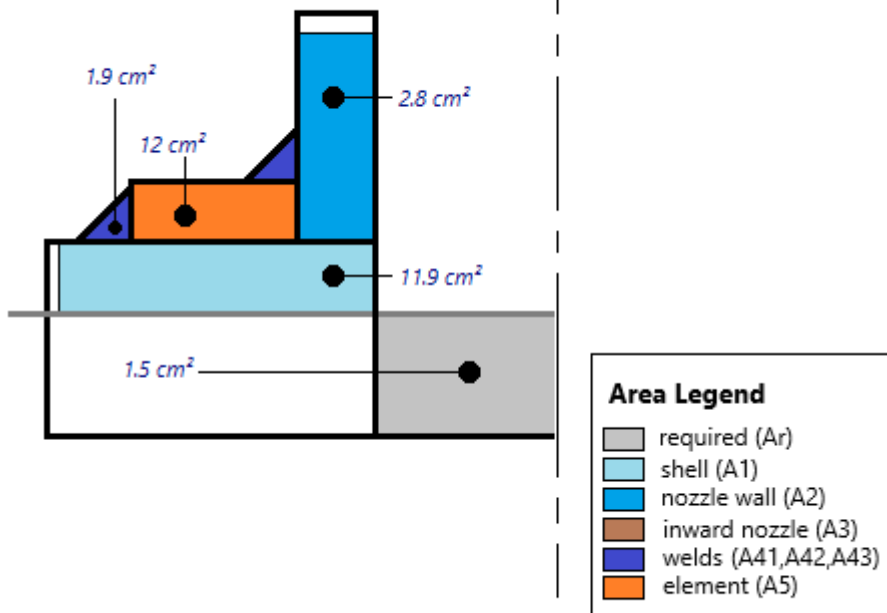


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1.532	NA	NA
Area in Shell	A1	11.942	NA	NA
Area in Nozzle Wall	A2	2.806	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.855	NA	NA
Area in Element	A5	12.000	NA	NA
TOTAL AREA AVAILABLE	Atot	28.603	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (152.6794 \cdot 1.0031 \cdot 1.0 + 2 \cdot 7.7978 \cdot 1.0031 \cdot 1.0 \cdot (1 - 1.0))$
 $= 1.532 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 152.679 (1.0 \cdot 8.825 - 1.0 \cdot 1.003) - 2 \cdot 7.798$
 $(1.0 \cdot 8.825 - 1.0 \cdot 1.0031) \cdot (1 - 1.0)$
 $= 11.942 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 22.06) \cdot (7.8 - 0.36) \cdot 0.8551$
 $= 2.806 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi\text{-}can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 10.0^2 \cdot 1.0$



= 1.855 cm²

Area Available in Element [A5]:

$$\begin{aligned} &= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, t_e)) \cdot fr4 \\ &= (268.275 - 168.275) \cdot 12.0 \cdot 1.0 \\ &= 12.000 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.5360 mm.
Wall Thickness per UG16(b),	tr16b = 5.6750 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 4.1781 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 5.6750 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 5.6750 mm.
Wall Thickness per table UG-45	tb3 = 9.3950 mm.

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[9.395, \max(5.675, 5.675)] \\ &= 5.6750 \text{ mm.} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(3.536, 5.675) \\ &= 5.6750 \text{ mm.} \end{aligned}$$

Available Nozzle Neck Thickness = 0.875 * 10.973 = 9.601 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 31.4,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 263.6 N./mm ²	Passed
Occasional	: 2.6,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 20.3,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.601, tr = 0.361, c = 3.175 mm., E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.056, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.601, tr = 0.361, c = 3.175 mm., E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.056, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.601, tr = 0.361, c = 3.175 mm., E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.056, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 12.0, tr = 1.003, c = 3.175 mm., E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.114, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.601, tr = 0.361, c = 3.175 mm., E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.056$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $5.56/51.00 = 0.109$

Weld Size Calculations, Description: TS IN NOZ T1

Intermediate Calc. for nozzle/shell Welds Tmin 7.7978 mm.
Intermediate Calc. for pad/shell Welds TminPad 8.8250 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.4585 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.
Pad Weld	$4.4125 = 0.5 \cdot t_{minPad}$	$7.0700 = 0.7 \cdot W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= (A_r - A_1) \cdot S_v$
 $= (1.5316 - 11.9424) \cdot 138$
 $= 0.00 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - C_{an} / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (2.8059 + 12.0 + 1.8551 - 0.0 \cdot 0.86) \cdot 138$
 $= 23444.62 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 15584. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 268.275 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 29056. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 168.275 \cdot 12.0 \cdot 0.74 \cdot 138$
 $= 33029. \text{ Kgf}$

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (S_{PEW} + S_{NGW}) = (29056 + 0) = 29056 \text{ Kgf}$



Summary of Failure Path Calculations:

Path 1-1 = 29056 Kgf, must exceed W = 0 Kgf or W1 = 23444 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 48 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 14.9003 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 176.9003 mm.

Input Echo, WRC297 Item 1, Description: TS IN NOZ T1 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.1750	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	490.000	mm.
Shell Thickness	12.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.1750	mm.
Nozzle Diameter	146.329	mm.
Nozzle Thickness	10.9728	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	268.275	mm.
Thickness of Reinforcing Pad	12.0000	mm.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUS tained loads.

Design Internal Pressure	Dp	5.56	bars
Radial Load	P	-407.91	Kgf
Circumferential Shear	Vc	611.87	Kgf
Longitudinal Shear	Vl	611.87	Kgf
Circumferential Moment	Mc	275.24	Kg-m.
Longitudinal Moment	Ml	366.98	Kg-m.
Torsional Moment	Mt	458.73	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))} \\
 &= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(258.587(12.0 - 3.175))} \\
 &= 325.918 \text{ mm.}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 517.175 mm.

Nozzle Outside Diameter (d) = 168.275 mm.
 Vessel Thickness used (T) = 20.825 mm.
 Nozzle Thickness used (t) = 7.798 mm.
 T / t = 2.671
 d / t = 21.580

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.621$$

Nr/P = 0.079
 Mr/P = 0.083
 M0/P = 0.071
 N0/P = 0.185
 MrD/Mc = 0.178
 NrDL/Mc = 0.128
 M0d/Mc = 0.170
 N0DL/Mc = 0.152
 MrD/Ml = 0.127
 NrDL/Ml = 0.091
 M0D/Ml = 0.122
 N0DL/Ml = 0.281

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-13	-13	13	13
Outplane Bending (ML)	-36	36	36	-36
Normal Pressure Stress	6	6	6	6
Outplane Stress Summary	-37	26	62	-17

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	4	-4	4	-4
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-37	37	37	-37
Inplane Pressure Stress	3	3	3	3
Inplane Stress Summary	-33	32	50	-33

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	42	36	63	34

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-37	37	37	-37
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	3	3	3	3
Outplane Stress Summary	-34	32	52	-31

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	4	-4	4	-4
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-39	39	39	-39
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	6	6	6	6
Inplane Stress Summary	-32	37	55	-31

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	3	3	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	37	39	60	37

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-13	-13	13	13
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	6	6	6
Hoop Stress Summary	-5	-5	21	22

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	17	-17	17	-17
Axial Membrane (Mc)	0	0	0	0



Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-23	-23	23	23
Axial Bending (ML)	-171	171	171	-171
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-174	134	216	-160
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	12	12	12	12

Shear Stress Summary	15	15	10	10
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	175	143	217	184
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-5	-5	5	5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	6	6	6

Hoop Stress Summary	2	2	13	14
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	17	-17	17	-17
Axial Membrane (Mc)	-17	-17	17	17
Axial Bending (Mc)	-179	179	179	-179
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-176	148	218	-174
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	12	12	12	12

Shear Stress Summary	10	10	15	15
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	179	149	219	191
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{252.587(12.0 - 3.175)}$$

$$= 324.078 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	505.175 mm.
Nozzle Outside Diameter	(d) =	268.275 mm.
Vessel Thickness used	(T) =	8.825 mm.
Nozzle Thickness used	(t) =	57.798 mm.
T / t	=	0.153
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.018$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.050
Mr/P	=	0.082
M0/P	=	0.025
N0/P	=	0.044
MrD/Mc	=	0.253
NrDL/Mc	=	0.096
M0d/Mc	=	0.076
N0DL/Mc	=	0.073
MrD/M1	=	0.064
NrDL/M1	=	0.053
M0D/M1	=	0.019
N0DL/M1	=	0.047

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-20	20	20	-20
Normal Pressure Stress	15	15	15	15
Outplane Stress Summary	-2	22	53	-1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	25	-25	25	-25
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-9	-9	9	9
Inplane Bending (ML)	-66	66	66	-66
Inplane Pressure Stress	7	7	7	7
Inplane Stress Summary	-39	41	111	-72

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0

Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	40	43	111	72

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	-9	-9	9	9
Outplane Bending (Mc)	-58	58	58	-58
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	7	7	7	7
Outplane Stress Summary	-50	51	85	-47

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	25	-25	25	-25
Inplane Membrane (Mc)	-12	-12	12	12
Inplane Bending (Mc)	-196	196	196	-196
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	15	15	15	15
Inplane Stress Summary	-165	177	252	-191

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	2	2	6	6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	165	177	252	191

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	6	6	6	6	6	6	6
Circ. Pl (SUS)	-12	-12	14	14	-4	-4	4	4

Circ. Q (SUS)	-33	33	39	-39	-35	35	43	-43
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-4	-4	4	4	-4	-4	6	6
Long. Q (SUS)	-33	33	41	-41	-34	34	40	-40
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
Pm+Pl (SUS)	6.2	6.2	20.1	20.1	3.6	3.6	10.6	10.6
Pm+Pl+Q (Total)	42.1	35.1	59.8	34.6	37.2	38.6	56.4	37.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	6.00	138.00	Passed
Pm+Pl (SUS)	20.08	207.00	Passed
Pm+Pl+Q (TOTAL)	59.76	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	15	15	15	15	15	15	15	15
Circ. Pl (SUS)	-6	-6	10	10	-10	-10	14	14
Circ. Q (SUS)	-13	13	27	-27	-171	171	221	-221
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-7	-7	11	11	-7	-7	11	11
Long. Q (SUS)	-41	41	91	-91	-51	51	65	-65
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0
Pm+Pl (SUS)	9.2	9.2	25.1	25.1	5.4	5.4	29.1	29.1
Pm+Pl+Q (Total)	41.7	42.2	109.2	73.1	166.1	176.1	250.1	192.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	15.00	138.00	Passed
Pm+Pl (SUS)	29.09	207.00	Passed
Pm+Pl+Q (TOTAL)	250.15	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of	Stress Values at
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Stress Int. Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	6	6	6	6	6	6	6
Circ. Pl (SUS)	-12	-12	14	14	-4	-4	6	6
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-22	-22	24	24	-16	-16	18	18
Long. Q (SUS)	-154	154	188	-188	-162	162	196	-196
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	12	12	12	12	12	12	12	12
Pm (SUS)	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
Pm+Pl (SUS)	19.3	19.3	27.5	27.5	15.5	15.5	21.4	21.4
Pm+Pl+Q (Total)	174.2	143.8	215.5	182.1	178.1	149.7	218.0	189.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	6.00	118.00	Passed
Pm+Pl (SUS)	27.53	177.00	Passed
Pm+Pl+Q (TOTAL)	217.95	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: TS OUT NOZ T2

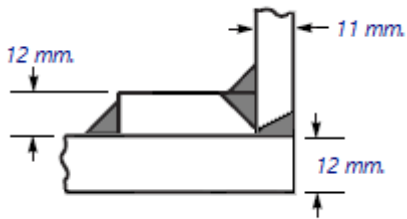
From: 20

Pressure for Reinforcement Calculations	P	5.516	bars
Temperature for Internal Pressure	Temp	77	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	490.00	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	c	3.1750	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		250.00	mm.
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.1750	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	268.2750	mm.
Thickness of Pad	te	12.0000	mm.
Weld leg size between Pad and Shell	Wp	10.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm.
Reinforcing Pad Width		50.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: TS OUT NOZ T2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	5.761 in.
Actual Thickness Used in Calculation	0.432 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.52 \cdot 248.175 / (138 \cdot 1.0 - 0.6 \cdot 5.52)$
 $= 0.9944$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.52 \cdot 76.34 / (118 \cdot 1.0 - 0.6 \cdot 5.52)$
 $= 0.3579$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	305.3588 mm.
Parallel to Vessel Wall, opening length	d	152.6794 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		22.0625 mm.

Weld Strength Reduction Factor [fr1]:
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

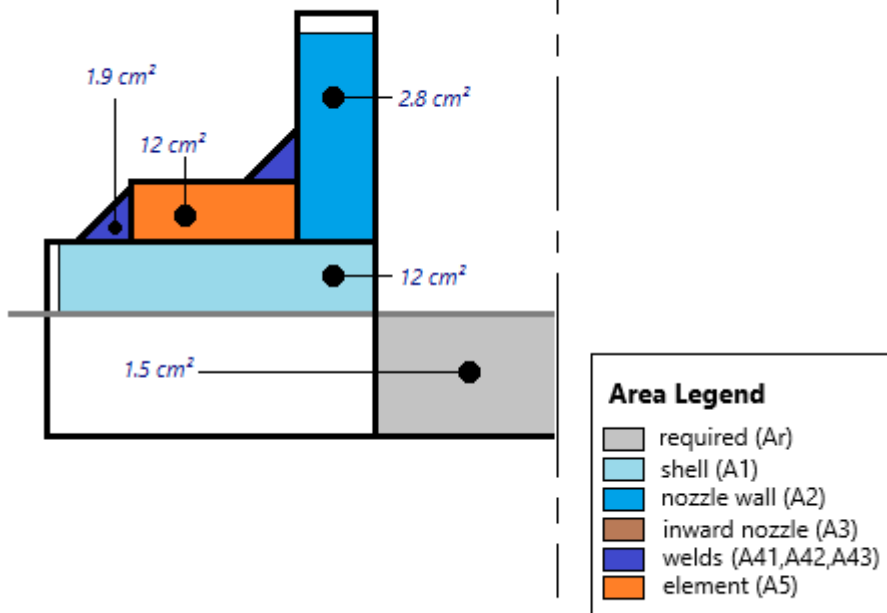


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1.518	NA	NA
Area in Shell	A1	11.956	NA	NA
Area in Nozzle Wall	A2	2.807	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.855	NA	NA
Area in Element	A5	12.000	NA	NA
TOTAL AREA AVAILABLE	Atot	28.618	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (152.6794 \cdot 0.9944 \cdot 1.0 + 2 \cdot 7.7978 \cdot 0.9944 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 1.518 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 152.679 (1.0 \cdot 8.825 - 1.0 \cdot 0.994) - 2 \cdot 7.798$$

$$(1.0 \cdot 8.825 - 1.0 \cdot 0.994) \cdot (1 - 1.0)$$

$$= 11.956 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 22.06) \cdot (7.8 - 0.36) \cdot 0.8551$$

$$= 2.807 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi\text{-}can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$

$$= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 10.0^2 \cdot 1.0$$



= 1.855 cm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))*(min(tp,Tlwp,te)) • fr4
= (268.275 - 168.275) • 12.0 • 1.0
= 12.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 3.5329 mm.
Wall Thickness per UG16(b), tr16b = 5.6750 mm.
Wall Thickness, shell/head, internal pressure trb1 = 4.1694 mm.
Wall Thickness tb1 = max(trb1, tr16b) = 5.6750 mm.
Wall Thickness tb2 = max(trb2, tr16b) = 5.6750 mm.
Wall Thickness per table UG-45 tb3 = 9.3950 mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[9.395, max(5.675, 5.675)]
= 5.6750 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(3.5329, 5.675)
= 5.6750 mm.

Available Nozzle Neck Thickness = 0.875 * 10.973 = 9.601 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	31.4,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	263.6 N./mm ²	Passed
Occasional	:	2.6,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	20.3,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.601, tr = 0.358, c = 3.175 mm., E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.056, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.601, tr = 0.358, c = 3.175 mm., E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.056, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.601, tr = 0.358, c = 3.175 mm., E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.056, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 12.0, tr = 0.994, c = 3.175 mm., E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.113, Temp. Reduction = 78 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.601, tr = 0.358, c = 3.175 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.056, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 5.52/51.00 = 0.108

Weld Size Calculations, Description: TS OUT NOZ T2

Intermediate Calc. for nozzle/shell Welds Tmin 7.7978 mm.
 Intermediate Calc. for pad/shell Welds TminPad 8.8250 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.4585 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.
Pad Weld	4.4125 = 0.5*TminPad	7.0700 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 = (Ar - A1) * Sv
 = (1.5183 - 11.9557) * 138
 = 0.00 Kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 = (A2+A5+A4-(Wi-Can/.707)²*fr2)*Sv
 = (2.8071 + 12.0 + 1.8551 - 0.0 * 0.86) * 138
 = 23446.27 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 = (π/2) * Dlo * Wo * 0.49 * Snw
 = (3.1416/2.0) * 168.275 * 10.0 * 0.49 * 118
 = 15584. Kgf

Shear, Pad Element Weld [Spew]:
 = (π/2) * DP * WP * 0.49 * SEW
 = (3.1416/2.0) * 268.275 * 10.0 * 0.49 * 138
 = 29056. Kgf

Tension, Pad Groove Weld [Tpgw]:
 = (π/2) * Dlo * Wgpn * 0.74 * Seg
 = (3.1416/2) * 168.275 * 12.0 * 0.74 * 138
 = 33029. Kgf

Strength of Failure Paths:

Failure Path [Path 1-1]:
 = (SPEW + SNGW) = (29056 + 0) = 29056 Kgf



Summary of Failure Path Calculations:

Path 1-1 = 29056 Kgf, must exceed W = 0 Kgf or W1 = 23446 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 48 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 14.9003 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 176.9003 mm.

Input Echo, WRC297 Item 1, Description: TS OUT NOZ T2 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.1750	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	490.000	mm.
Shell Thickness	12.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.1750	mm.
Nozzle Diameter	146.329	mm.
Nozzle Thickness	10.9728	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	268.275	mm.
Thickness of Reinforcing Pad	12.0000	mm.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	5.52	bars
Radial Load	P	-407.91	Kgf
Circumferential Shear	Vc	611.87	Kgf
Longitudinal Shear	Vl	611.87	Kgf
Circumferential Moment	Mc	275.24	Kg-m.
Longitudinal Moment	Ml	366.98	Kg-m.
Torsional Moment	Mt	458.73	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))} \\
 &= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(258.587(12.0 - 3.175))} \\
 &= 325.918 \text{ mm.}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 517.175 mm.

Nozzle Outside Diameter (d) = 168.275 mm.
 Vessel Thickness used (T) = 20.825 mm.
 Nozzle Thickness used (t) = 7.798 mm.
 T / t = 2.671
 d / t = 21.580

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.621$$

Nr/P = 0.079
 Mr/P = 0.083
 M0/P = 0.071
 N0/P = 0.185
 MrD/Mc = 0.178
 NrDL/Mc = 0.128
 M0d/Mc = 0.170
 N0DL/Mc = 0.152
 MrD/Ml = 0.127
 NrDL/Ml = 0.091
 M0D/Ml = 0.122
 N0DL/Ml = 0.281

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-13	-13	13	13
Outplane Bending (ML)	-36	36	36	-36
Normal Pressure Stress	6	6	6	6
Outplane Stress Summary	-38	26	61	-17

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	4	-4	4	-4
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-37	37	37	-37
Inplane Pressure Stress	3	3	3	3
Inplane Stress Summary	-33	32	50	-33

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	42	36	63	34

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-37	37	37	-37
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	3	3	3	3
Outplane Stress Summary	-34	32	52	-31

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	4	-4	4	-4
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-39	39	39	-39
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	6	6	6	6
Inplane Stress Summary	-32	37	55	-31

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	3	3	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	37	39	60	37

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-13	-13	13	13
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	6	6	6
Hoop Stress Summary	-5	-5	21	22

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	17	-17	17	-17
Axial Membrane (Mc)	0	0	0	0



Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-23	-23	23	23
Axial Bending (ML)	-171	171	171	-171
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-174	134	216	-160
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	12	12	12	12

Shear Stress Summary	15	15	10	10
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	175	143	217	184
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-5	-5	5	5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	6	6	6

Hoop Stress Summary	2	2	13	14
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	17	-17	17	-17
Axial Membrane (Mc)	-17	-17	17	17
Axial Bending (Mc)	-179	179	179	-179
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-176	148	218	-174
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	12	12	12	12

Shear Stress Summary	10	10	15	15
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	179	149	219	191
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**Stress Computations at the Edge of the Reinforcing Pad:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{252.587(12.0 - 3.175)}$$

$$= 324.078 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	505.175 mm.
Nozzle Outside Diameter	(d) =	268.275 mm.
Vessel Thickness used	(T) =	8.825 mm.
Nozzle Thickness used	(t) =	57.798 mm.
T / t	=	0.153
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.018$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.050
Mr/P	=	0.082
M0/P	=	0.025
N0/P	=	0.044
MrD/Mc	=	0.253
NrDL/Mc	=	0.096
M0d/Mc	=	0.076
N0DL/Mc	=	0.073
MrD/M1	=	0.064
NrDL/M1	=	0.053
M0D/M1	=	0.019
N0DL/M1	=	0.047

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-20	20	20	-20
Normal Pressure Stress	15	15	15	15
Outplane Stress Summary	-3	22	53	-1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	25	-25	25	-25
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-9	-9	9	9
Inplane Bending (ML)	-66	66	66	-66
Inplane Pressure Stress	7	7	7	7
Inplane Stress Summary	-39	41	111	-72

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0



Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	40	43	111	72

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	-9	-9	9	9
Outplane Bending (Mc)	-58	58	58	-58
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	7	7	7	7
Outplane Stress Summary	-50	51	85	-47

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	25	-25	25	-25
Inplane Membrane (Mc)	-12	-12	12	12
Inplane Bending (Mc)	-196	196	196	-196
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	15	15	15	15
Inplane Stress Summary	-165	177	252	-191

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	2	2	6	6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	165	177	252	191

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	6	6	6	6	6	6	6
Circ. Pl (SUS)	-12	-12	14	14	-4	-4	4	4

Circ. Q (SUS)	-33	33	39	-39	-35	35	43	-43
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-4	-4	4	4	-4	-4	6	6
Long. Q (SUS)	-33	33	41	-41	-34	34	40	-40
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
Pm+Pl (SUS)	6.2	6.2	20.1	20.1	3.6	3.6	10.6	10.6
Pm+Pl+Q (Total)	42.1	35.1	59.8	34.6	37.2	38.6	56.4	37.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	6.00	138.00	Passed
Pm+Pl (SUS)	20.08	207.00	Passed
Pm+Pl+Q (TOTAL)	59.76	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	15	15	15	15	15	15	15	15
Circ. Pl (SUS)	-6	-6	10	10	-10	-10	14	14
Circ. Q (SUS)	-13	13	27	-27	-171	171	221	-221
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-7	-7	11	11	-7	-7	11	11
Long. Q (SUS)	-41	41	91	-91	-51	51	65	-65
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0
Pm+Pl (SUS)	9.2	9.2	25.1	25.1	5.4	5.4	29.1	29.1
Pm+Pl+Q (Total)	41.7	42.2	109.2	73.1	166.1	176.1	250.1	192.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	15.00	138.00	Passed
Pm+Pl (SUS)	29.09	207.00	Passed
Pm+Pl+Q (TOTAL)	250.15	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of	Stress Values at
---------	------------------



Stress Int. Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	6	6	6	6	6	6	6
Circ. Pl (SUS)	-12	-12	14	14	-4	-4	6	6
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-22	-22	24	24	-16	-16	18	18
Long. Q (SUS)	-154	154	188	-188	-162	162	196	-196
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	12	12	12	12	12	12	12	12
Pm (SUS)	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
Pm+Pl (SUS)	19.3	19.3	27.5	27.5	15.5	15.5	21.4	21.4
Pm+Pl+Q (Total)	174.2	143.8	215.5	182.1	178.1	149.7	218.0	189.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	6.00	118.00	Passed
Pm+Pl (SUS)	27.53	177.00	Passed
Pm+Pl+Q (TOTAL)	217.95	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: SH IN NOZ S1

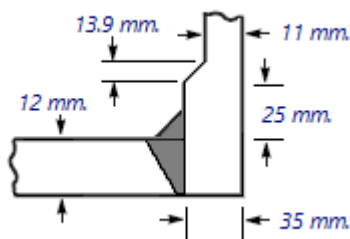
From: 50

Pressure for Reinforcement Calculations	P	17.237	bars
Temperature for Internal Pressure	Temp	132	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	490.00	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	c	3.1750	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		6510.50	mm.
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	25.0000	mm.
Height of Beveled Transition	L'	13.8721	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		35.0000	mm.
Corrosion Allowance	can	3.1750	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		10.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: SH IN NOZ S1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 5.761 in.
Actual Thickness Used in Calculation 0.432 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $17.24 \cdot 248.175 / (138 \cdot 1.0 - 0.6 \cdot 17.24)$
= 3.1235 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $17.24 \cdot 76.34 / (118 \cdot 1.0 - 0.6 \cdot 17.24)$
= 1.1251 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= $24.027 / 0.5773$
= 41.6164 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

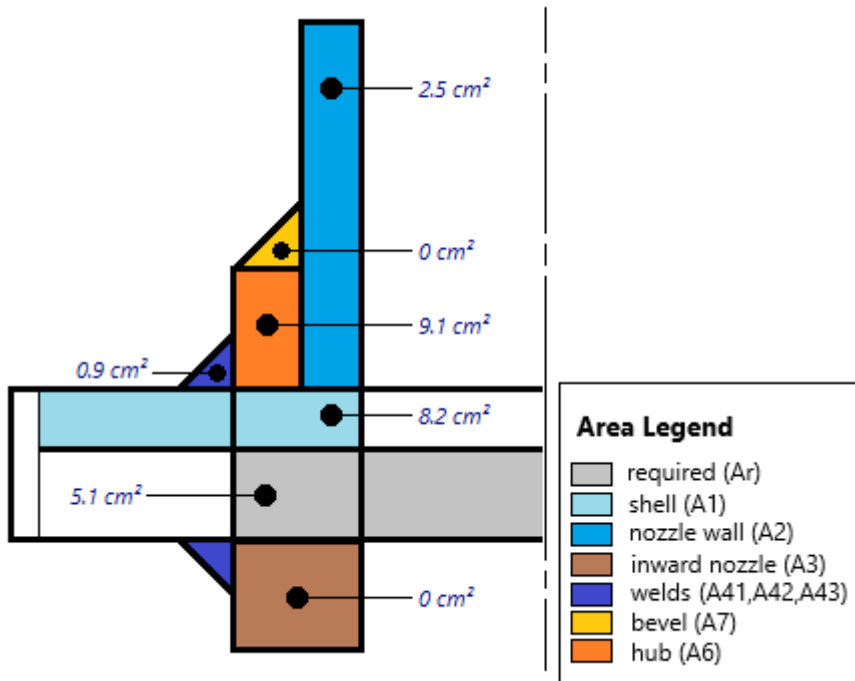
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	305.3588	mm.
Parallel to Vessel Wall, opening length	d	152.6794	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.0625	mm.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 118.0 / 138.0)$
= 0.855

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 118.0 / 138.0)$
= 0.855

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(0.855, 1.0)$
= 0.855



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	5.057	NA	NA
Area in Shell	A1	8.179	NA	NA
Area in Nozzle Wall	A2	2.518	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	9.065	NA	NA
TOTAL AREA AVAILABLE	Atot	20.617	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (152.6794 \cdot 3.1235 \cdot 1.0 + 2 \cdot 31.825 \cdot 3.1235 \cdot 1.0 \cdot (1 - 0.86))$$

$$= 5.057 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 152.679(1.0 \cdot 8.825 - 1.0 \cdot 3.123) - 2 \cdot 31.825$$

$$(1.0 \cdot 8.825 - 1.0 \cdot 3.1235) \cdot (1 - 0.855)$$

$$= 8.179 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 22.06)(7.8 - 1.13)0.8551$$

$$= 2.518 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$$

$$= 0.855 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2 \\ &= (2 \cdot \min(22.1, 150.0, 25.0)) \cdot (35.0 - 11.0) \cdot 0.855 \\ &= 9.065 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

$$\begin{aligned} \text{Wall Thickness for Internal/External pressures} & \quad ta = 4.3001 \text{ mm.} \\ \text{Wall Thickness per UG16(b),} & \quad tr16b = 4.6750 \text{ mm.} \\ \text{Wall Thickness, shell/head, internal pressure} & \quad trb1 = 6.2985 \text{ mm.} \\ \text{Wall Thickness} & \quad tb1 = \max(trb1, tr16b) = 6.2985 \text{ mm.} \\ \text{Wall Thickness} & \quad tb2 = \max(trb2, tr16b) = 4.6750 \text{ mm.} \\ \text{Wall Thickness per table UG-45} & \quad tb3 = 9.3950 \text{ mm.} \end{aligned}$$

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[9.395, \max(6.2985, 4.675)] \\ &= 6.2985 \text{ mm.} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(4.3001, 6.2985) \\ &= 6.2985 \text{ mm.} \end{aligned}$$

Available Nozzle Neck Thickness = $0.875 \cdot 10.973 = 9.601 \text{ mm.}$ -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	36.8,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	258.1 N./mm ²	Passed
Occasional	:	8.0,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	20.3,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.601, tr = 1.125, c = 3.175 mm., E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.175$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 12.0, tr = 3.123, c = 3.175 mm., E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.354$, Temp. Reduction = 75 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $17.24/51.00 = 0.338$

Weld Size Calculations, Description: SH IN NOZ S1

Intermediate Calc. for nozzle/shell Welds Tmin 8.8250 mm.



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (5.0571 - 8.1791 + 2 \cdot 31.825 \cdot 0.8551 \cdot (1.0 \cdot 8.825 - 3.1235)) \cdot 138)$$

$$= \max(0, -26.67) \text{ Kg}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (11.5831 + 0.8551 - 0.0 \cdot 0.86) \cdot 138$$

$$= 17502.46 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (11.5831 + 0.0 + 0.8551 + (4.803)) \cdot 138$$

$$= 24261.09 \text{ Kg}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (11.5831 + 0.0 + 0.8551 + 0.0 + (4.803)) \cdot 138$$

$$= 24261.09 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 216.3294 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 20034. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 92.2522) \cdot (35.0 - 3.175) \cdot 0.7 \cdot 118$$

$$= 77685. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 216.3294 \cdot (12.0 - 3.175) \cdot 0.74 \cdot 138$$

$$= 31227. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (20034 + 77685) = 97720 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (20034 + 0 + 31227 + 0) = 51261 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (20034 + 31227 + 0) = 51261 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 97719	Kgf, must exceed W = 0 Kg	or W1 = 17502 Kg
Path 2-2 = 51261	Kgf, must exceed W = 0 Kg	or W2 = 24261 Kg
Path 3-3 = 51261	Kgf, must exceed W = 0 Kg	or W3 = 24261 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 44 bars



The Drop for this Nozzle is : 25.1696 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 187.1696 mm.

Input Echo, WRC297 Item 1, Description: SH IN NOZ S1 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.1750	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	490.000	mm.
Shell Thickness	12.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.1750	mm.
Nozzle Diameter	146.329	mm.
Nozzle Thickness	35.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.*

Design Internal Pressure	Dp	17.24	bars
Radial Load	P	-407.91	Kgf
Circumferential Shear	Vc	611.87	Kgf
Longitudinal Shear	Vl	611.87	Kgf
Circumferential Moment	Mc	275.24	Kg-m.
Longitudinal Moment	Ml	366.98	Kg-m.
Torsional Moment	Mt	458.73	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (6.181) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 216.329 + 2 \cdot 1.65 \cdot \sqrt{(252.587(12.0 - 3.175))}$
 $= 372.133 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	505.175	mm.
Nozzle Outside Diameter	(d) =	216.329	mm.
Vessel Thickness used	(T) =	8.825	mm.
Nozzle Thickness used	(t) =	31.825	mm.
T / t	=	0.277	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.240$$

Nr/P	=	0.057
Mr/P	=	0.098

MØ/P = 0.031
 NØ/P = 0.066
 MrD/Mc = 0.262
 NrDL/Mc = 0.093
 MØd/Mc = 0.079
 NØDL/Mc = 0.086
 MrD/Ml = 0.082
 NrDL/Ml = 0.056
 MØD/Ml = 0.023
 NØDL/Ml = 0.076

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-16	-16	16	16
Outplane Bending (ML)	-29	29	29	-29
Normal Pressure Stress	47	49	47	49
Outplane Stress Summary	14	57	106	29

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	30	-30	30	-30
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-12	-12	12	12
Inplane Bending (ML)	-104	104	104	-104
Inplane Pressure Stress	23	23	23	23
Inplane Stress Summary	-59	89	173	-96

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	6	6	6	6
Shear Stress Summary	8	8	4	4

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	76	91	174	126

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	-13	-13	13	13
Outplane Bending (Mc)	-75	75	75	-75
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0



Normal Pressure Stress	23	23	23	23
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Outplane Stress Summary	-52	79	126	-44
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	30	-30	30	-30
Inplane Membrane (Mc)	-14	-14	14	14
Inplane Bending (Mc)	-252	252	252	-252
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	47	49	47	49

Inplane Stress Summary	-186	259	347	-215
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	6	6	6	6

Shear Stress Summary	4	4	8	8
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	186	259	348	215
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-16	-16	16	16
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	6	4	6

Hoop Stress Summary	-8	-6	23	25
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-4	-4	4	4
Axial Bending (ML)	-5	5	5	-5
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-6	1	14	0
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	2	2	1	1
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	10	9	23	25
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-13	-13	13	13
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	6	4	6

Hoop Stress Summary	-6	-4	21	23
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-3	-3	3	3
Axial Bending (Mc)	-15	15	15	-15
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-15	12	23	-11
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	1	1	2	2
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	15	17	25	35
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	47	49	47	49	47	49	47	49
Circ. Pl (SUS)	-13	-13	19	19	-12	-12	16	16
Circ. Q (SUS)	-20	20	38	-38	-222	222	282	-282

Long. Pm (SUS)	23	23	23	23	23	23	23	23
Long. Pl (SUS)	-10	-10	14	14	-10	-10	16	16
Long. Q (SUS)	-74	74	134	-134	-66	66	84	-84
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	47.0	49.0	47.0	49.0	47.0	49.0	47.0	49.0
Pm+Pl (SUS)	34.0	36.0	66.0	68.0	35.0	37.0	63.0	65.0
Pm+Pl+Q (Total)	76.3	88.5	171.4	127.4	187.2	259.1	345.2	217.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	49.00	138.00	Passed
Pm+Pl (SUS)	68.03	207.00	Passed
Pm+Pl+Q (TOTAL)	345.22	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	6	4	6	4	6	4	6
Circ. Pl (SUS)	-13	-13	19	19	-10	-10	16	16
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-4	-4	4	4	-3	-3	3	3
Long. Q (SUS)	-4	4	6	-6	-14	14	16	-16
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	4.0	6.0	4.0	6.0	4.0	6.0	4.0	6.0
Pm+Pl (SUS)	9.0	7.0	23.0	25.0	6.0	4.0	20.0	22.0
Pm+Pl+Q (Total)	9.3	9.2	23.1	25.1	15.1	17.1	21.6	33.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	6.00	118.00	Passed
Pm+Pl (SUS)	25.00	177.00	Passed
Pm+Pl+Q (TOTAL)	33.06	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: SH OUT NOZ S2

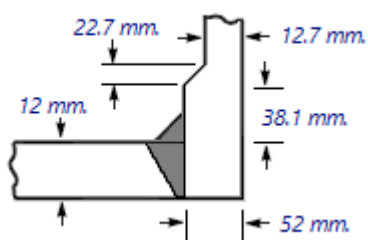
From: 50

Pressure for Reinforcement Calculations	P	17.285	bars
Temperature for Internal Pressure	Temp	132	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	490.00	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	c	3.1750	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		914.50	mm.
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	38.0599	mm.
Height of Beveled Transition	L'	22.6899	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		52.0000	mm.
Corrosion Allowance	can	3.1750	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		10.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: SH OUT NOZ S2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 7.625 in.
Actual Thickness Used in Calculation 0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $17.29 \cdot 248.175 / (138 \cdot 1.0 - 0.6 \cdot 17.29)$
= 3.1323 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $17.29 \cdot 100.01 / (118 \cdot 1.0 - 0.6 \cdot 17.29)$
= 1.4781 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= $39.3 / 0.5773$
= 68.0696 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

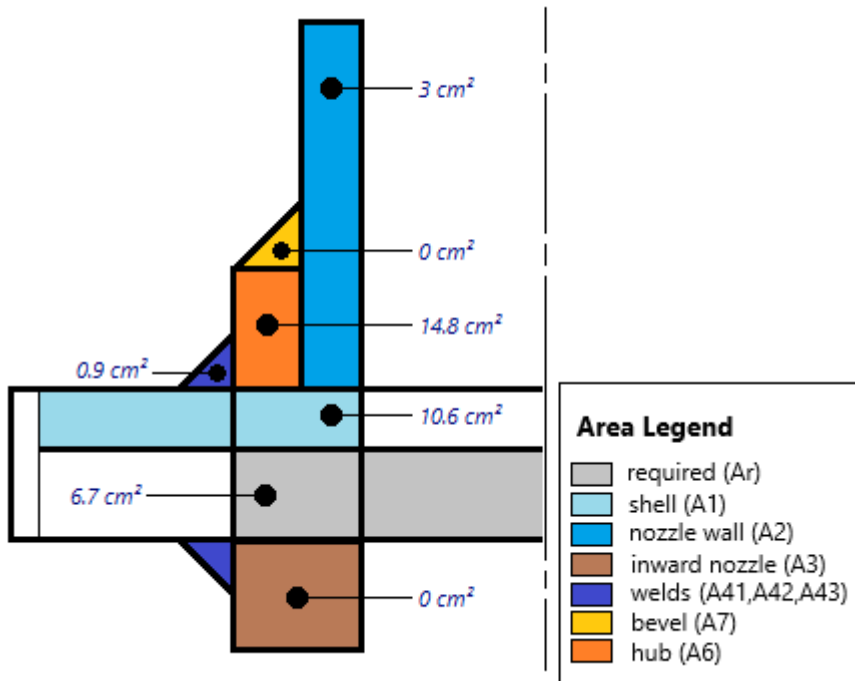
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	400.0500	mm.
Parallel to Vessel Wall, opening length	d	200.0250	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.0625	mm.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 118.0 / 138.0)$
= 0.855

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 118.0 / 138.0)$
= 0.855

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(0.855, 1.0)$
= 0.855



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	6.709	NA	NA
Area in Shell	A1	10.581	NA	NA
Area in Nozzle Wall	A2	3.036	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	14.828	NA	NA
TOTAL AREA AVAILABLE	Atot	29.300	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (200.025 \cdot 3.1323 \cdot 1.0 + 2 \cdot 48.825 \cdot 3.1323 \cdot 1.0 \cdot (1 - 0.86))$
 $= 6.709 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 200.025(1.0 \cdot 8.825 - 1.0 \cdot 3.132) - 2 \cdot 48.825$
 $(1.0 \cdot 8.825 - 1.0 \cdot 3.1323) \cdot (1 - 0.855)$
 $= 10.581 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 22.06)(9.52 - 1.48)0.8551$
 $= 3.036 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$
 $= 0.855 \text{ cm}^2$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(22.1, 150.0, 38.1)) \cdot (52.0 - 12.7) \cdot 0.855$$

$$= 14.828 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 4.6531 \text{ mm.}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.6750 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 6.3073 \text{ mm.}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 6.3073 \text{ mm.}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 4.6750 \text{ mm.}$
Wall Thickness per table UG-45	$tb_3 = 10.3350 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[10.335, \max(6.3073, 4.675)]$$

$$= 6.3073 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(4.6531, 6.3073)$$

$$= 6.3073 \text{ mm.}$$

Available Nozzle Neck Thickness = $0.875 \cdot 12.700 = 11.113 \text{ mm.} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 33.2,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 261.8 N./mm ²	Passed
Occasional	: 8.7,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 17.1,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 11.113$, $t_r = 1.478$, $c = 3.175 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.186$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-104 $^\circ\text{C}$
Min Metal Temp. w/o impact per UG-20(f)	-29 $^\circ\text{C}$

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 3.132$, $c = 3.175 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.355$, Temp. Reduction = $74 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-48 $^\circ\text{C}$
Min Metal Temp. w/o impact per UG-20(f)	-29 $^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : -48 $^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 $^\circ\text{C}$
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 $^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $17.29/51.00 = 0.339$

Weld Size Calculations, Description: SH OUT NOZ S2

Intermediate Calc. for nozzle/shell Welds $T_{min} = 8.8250 \text{ mm.}$



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (6.7086 - 10.5812 + 2 • 48.825 • 0.8551 •
(1.0 • 8.825 - 3.1323))138)
= 1239.22 Kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (17.864 + 0.8551 - 0.0 • 0.86) • 138
= 26340.69 Kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (17.864 + 0.0 + 0.8551 + (7.3687)) • 138
= 36709.59 Kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (17.864 + 0.0 + 0.8551 + 0.0 + (7.3687)) • 138
= 36709.59 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 297.675 • 10.0 • 0.49 • 118
= 27568. Kgf

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 124.425) • (52.0 - 3.175) • 0.7 • 118
= 160747. Kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.1416/2.0) • 297.675 • (12.0 - 3.175) • 0.74 • 138
= 42969. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (27568 + 160747) = 188315 Kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (27568 + 0 + 42969 + 0) = 70537 Kgf

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (27568 + 42969 + 0) = 70537 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 188315 Kgf, must exceed W = 1239 Kgf	or W1 = 26340 Kgf
Path 2-2 = 70536 Kgf, must exceed W = 1239 Kgf	or W2 = 36709 Kgf
Path 3-3 = 70536 Kgf, must exceed W = 1239 Kgf	or W3 = 36709 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 46 bars



The Drop for this Nozzle is : 50.3917 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 212.3917 mm.

Input Echo, WRC297 Item 1, Description: SH OUT NOZ S2 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.1750	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	490.000	mm.
Shell Thickness	12.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.1750	mm.
Nozzle Diameter	193.675	mm.
Nozzle Thickness	52.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUS tained loads.

Design Internal Pressure	Dp	17.29	bars
Radial Load	P	-550.68	Kgf
Circumferential Shear	Vc	815.83	Kgf
Longitudinal Shear	Vl	815.83	Kgf
Circumferential Moment	Mc	489.31	Kg-m.
Longitudinal Moment	Ml	652.41	Kg-m.
Torsional Moment	Mt	815.51	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Dv (.579) is > 0.5
check the Limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (5.725) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 297.675 + 2 \cdot 1.65 \cdot \sqrt{(252.587(12.0 - 3.175))}$$

$$= 453.479 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	505.175	mm.
Nozzle Outside Diameter	(d) =	297.675	mm.
Vessel Thickness used	(T) =	8.825	mm.
Nozzle Thickness used	(t) =	48.825	mm.
T / t	=	0.181	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.458$$



Nr/P = 0.045
 Mr/P = 0.074
 M0/P = 0.023
 N0/P = 0.046
 MrD/Mc = 0.250
 NrDL/Mc = 0.089
 M0d/Mc = 0.075
 N0DL/Mc = 0.078
 MrD/Ml = 0.061
 NrDL/Ml = 0.044
 M0D/Ml = 0.019
 N0DL/Ml = 0.045

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-12	-12	12	12
Outplane Bending (ML)	-31	31	31	-31
Normal Pressure Stress	47	49	47	49
Outplane Stress Summary	17	61	104	24

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	30	-30	30	-30
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-12	-12	12	12
Inplane Bending (ML)	-101	101	101	-101
Inplane Pressure Stress	23	23	23	23
Inplane Stress Summary	-55	85	171	-92

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	6	6	6	6
Shear Stress Summary	8	8	4	4

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	74	87	171	117

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	-16	-16	16	16
Outplane Bending (Mc)	-92	92	92	-92



Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	23	23	23	23
Outplane Stress Summary	-72	94	145	-59

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	30	-30	30	-30
Inplane Membrane (Mc)	-18	-18	18	18
Inplane Bending (Mc)	-309	309	309	-309
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	47	49	47	49
Inplane Stress Summary	-246	313	410	-269

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	6	6	6	6
Shear Stress Summary	4	4	8	8

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	246	313	410	270

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-12	-12	12	12
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	5	3	5
Hoop Stress Summary	-5	-3	19	21

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-2	2	2	-2
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-2	0	7	2

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	1	1	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	6	4	19	21

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-16	-16	16	16
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	5	3	5
Hoop Stress Summary	-9	-7	22	24

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-8	8	8	-8
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-8	7	13	-4

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	9	14	22	29

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	47	49	47	49	47	49	47	49

Circ. P1 (SUS)	-9	-9	15	15	-15	-15	21	21
Circ. Q (SUS)	-22	22	40	-40	-279	279	339	-339
Long. Pm (SUS)	23	23	23	23	23	23	23	23
Long. P1 (SUS)	-9	-9	15	15	-13	-13	19	19
Long. Q (SUS)	-71	71	131	-131	-83	83	101	-101
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	47.0	49.0	47.0	49.0	47.0	49.0	47.0	49.0
Pm+P1 (SUS)	38.0	40.0	62.0	64.0	32.0	34.0	68.0	70.0
Pm+P1+Q (Total)	74.3	87.0	169.4	117.4	247.1	313.1	407.2	269.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	49.00	138.00	Passed
Pm+P1 (SUS)	70.04	207.00	Passed
Pm+P1+Q (TOTAL)	407.19	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	5	3	5	3	5	3	5
Circ. P1 (SUS)	-9	-9	15	15	-13	-13	19	19
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. P1 (SUS)	-3	-3	3	3	-2	-2	2	2
Long. Q (SUS)	-2	2	2	-2	-8	8	8	-8
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	3.0	5.0	3.0	5.0	3.0	5.0	3.0	5.0
Pm+P1 (SUS)	6.0	4.0	18.0	20.0	10.0	8.0	22.0	24.0
Pm+P1+Q (Total)	6.4	4.5	18.1	20.1	10.6	15.1	22.1	29.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	118.00	Passed
Pm+P1 (SUS)	24.00	177.00	Passed
Pm+P1+Q (TOTAL)	29.07	353.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



Input Echo, Tubesheet Number 1, Description: U-Tubesheet

Shell Data:

Main Shell Description: Main shell

Shell Maximum Design Pressure	Psd,max	25.51 bars
Shell Maximum Operating Pressure	Psox,max	15.32 bars
Shell Minimum Operating Pressure	Psox,min	0.00 bars
Shell Thickness	ts	12.0000 mm.
Shell Internal Corrosion Allowance	cas	3.1750 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	490.000 mm.
Shell Temperature for Internal Pressure	Ts	126.67 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: CH.SHELL

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	5.52 bars
Channel Maximum Operating Pressure	Ptox,max	5.00 bars
Channel Minimum Operating Pressure	Ptox,min	0.00 bars
Channel Thickness	tc	12.0000 mm.
Channel Internal Corrosion Allowance	cac	3.1750 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	490.000 mm.
Channel Design Temperature	TEMPC	76.67 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	160
Tube Wall Thickness	et	2.7700 mm.
Tube Outside Diameter	D	19.0500 mm.
Total Straight Tube Length	Lt	6093.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		6041.00 mm.
Design Temperature of the Tubes		126.67 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	160.80 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm.
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.7700 mm.
Groove Weld Leg	ag	2.7700 mm.
Tube-Tubesheet Joint Weld Type	Full	Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	228.665 mm.
Largest Center-to-Center Tube Distance	Ul	35.0000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	6.0000 mm.

Tubesheet Data:



Tubesheet TYPE: U-tube, Gasketed both Sides, Conf. d

Tubesheet Design Metal Temperature T 126.67 °C
 Tubesheet Material SA-266 4

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03017
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	52.0000 mm.
Tubesheet Corr. Allowance (Shell side)	Cats	3.1470 mm.
Tubesheet Corr. Allowance (Channel side)	Catc	3.1750 mm.
Tubesheet Outside Diameter	A	622.000 mm.

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		No
Step 1 Diameter on the Tubeside	dt1	521.00 mm.
Step 1 Depth on the Tubeside	ht1	6.00 mm.
Step 2 Diameter on the Tubeside	dt2	487.00 mm.
Step 2 Depth on the Tubeside	ht2	6.00 mm.
Is the Tubesheet Flat on Shellside?		No
Step 1 Diameter on the Shellside	ds1	521.00 mm.
Step 1 Depth on the Shellside	hs1	6.00 mm.
Step 2 Diameter on the Shellside	ds2	487.00 mm.
Step 2 Depth on the Shellside	hs2	6.00 mm.
Calculated Tubesheet Diameter per Div 2, 4.18.4		487.00 mm.

Note: Tubesheet diameter is now: 487.000 mm. Div 2, 4.18.4.

Dimension G for the Channel Side	Gc	508.000 mm.
Area of the Untubed Lanes	AL	253.0 cm ²

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Both
Flange Outside Diameter	A	622.000 mm.
Flange Inside Diameter	B	490.000 mm.
Flange Face Outside Diameter	Fod	524.000 mm.
Flange Face Inside Diameter	Fid	490.000 mm.
Gasket Outside Diameter	Go	518.000 mm.
Gasket Inside Diameter	Gi	498.000 mm.
Small end Hub thk.	g0	12.0000 mm.
Large end Hub thk.	g1	14.0000 mm.
Gasket Factor,	m	3.75
Gasket Design Seating Stress	y	62.06 N./mm ²
Flange Facing Sketch	Code Sketch 1a	
Gasket Thickness	tg	4.5000 mm.
Full face Gasket Flange Option	Program Selects	
Length of Partition Gasket	lp	498.000 mm.
Width of Partition Gasket	wp	10.0000 mm.
Partition Gasket Factor,	mPart	3.7500
Partition Gasket Design Seating Stress	yPart	52.40 N./mm ²
Partition Gasket Facing Sketch	Code Sketch 1a	
Partition Gasket Column for Gasket Seating	Code Column II	

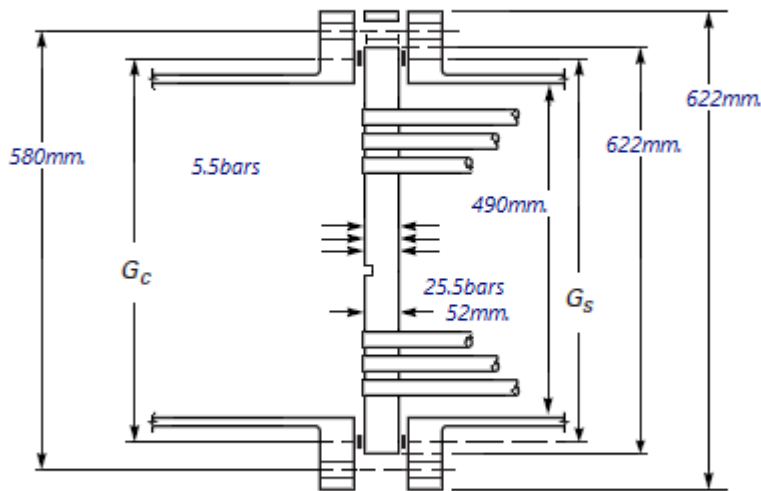
Bolting Information:

Diameter of Bolt Circle	C	580.000 mm.
Nominal Bolt Diameter	dB	19.0500 mm.
Thread Series	TEMA Thread Series	
Number of Bolts	n	24
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm.

Alternate Flange Operating Bolt Load, Wm1 73155.59 Kgf
 Alternate Flange Seating Bolt Load, Wm2 63798.16 Kgf
 Alternate Flange Design Bolt Load, W 82012.02 Kgf

Tubesheet Integral with None
 Tubesheet Extended as Flange Yes
 Thickness of Extended Portion of Tubesheet Tf 32.0000 mm.
 Is Bolt Load Transferred to the Tubesheet Yes

Tubesheet Gasketed w/Shell and Channel



Configuration d:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	126.7 °C	0.19650E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	76.7 °C	0.19952E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubsht. Design Temp.	126.7 °C	0.19650E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	126.7 °C	0.19650E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (5.52 \cdot (19.05/2 - 0)) / (92.4 \cdot 1.0 + 0.4 \cdot 5.52)$
 $= 0.0567 + 0.0000 = 0.0567 \text{ mm.}$

Tube Required Thickness under External Pressure (Shellside pressure) :



External Pressure Chart CS-1 at 126.67 °C
Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.7700	19.05	6269.67	6.88	50.0000	0.0232575	95.84

EMAWP = $(2.167 / (D/T) - 0.0833) * B = 222.1421$ bars

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.5129	19.05	6269.67	37.14	50.0000	0.0007975	71.07

EMAWP = $(4 * B) / (3 * (D/T)) = (4 * 71.0702) / (3 * 37.14) = 25.5128$ bars

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.5129 mm.
Allowable Internal Pressure at Corroded thickness	304.06 bars
Required Internal Design Pressure	5.52 bars
Allowable External Pressure at Corroded thickness	222.14 bars
Required External Design Pressure	25.51 bars
Required Thickness due to Shell Side pressure	0.5129 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	10.000 mm.
Basic Gasket Width,	$b_0 = N / 2$	5.000 mm.
Effective Gasket Width,	$b = b_0$	5.000 mm.
Gasket Reaction Diameter,	$G = (G_o + G_i) / 2$	508.000 mm.

Results for ASME U-tube Tubesheet Calculations for Configuration d:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:
 $= 2 * r_o + d_t$
 $= 2 * 228.665 + 19.05$
 $= 476.380$ mm.

Basic ligament efficiency for shear [μ]:
 $= (p - d_t) / p$
 $= (31.75 - 19.05) / 31.75$
 $= 0.400$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:
 $= G_s / D_o$ (Configurations d, e, f)
 $= 508.0 / 476.38$
 $= 1.0664$

Ratio [ρ_c]:
 $= G_c / D_o$ (Configurations d)
 $= 508.0 / 476.38$
 $= 1.0664$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:
 $= l_{tx} / h$
 $= 0 / 52.0$
 $= 0.000$ (must be $0 \leq \rho \leq 1$)



Effective tube hole Diameter [d*]:

$$\begin{aligned} &= \max(dt - 2tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2tt) \\ &= \max(19.05 - 2 \cdot 2.77 (0.19650E+09/.19650E+09) \cdot \\ &\quad (92/137) \cdot (0); 19.05 - 2 \cdot 2.77) \\ &= 19.0500 \text{ mm.} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(A/4 \cdot Do \cdot p) / (\pi \cdot Do^2)) } \\ &= 31.75 / \sqrt{ (1 - 4 \cdot \min(252.98 ; 4 \cdot 476.38 \cdot 31.75) / (\pi \cdot 476.38^2)) } \\ &= 34.2755 \text{ mm.} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (34.2755 - 19.05) / 34.2755 \\ &= 0.4442 \end{aligned}$$

$$\begin{aligned} h/p &= 1.637795 ; \mu^* = 0.444209 ; E^* = 95607992 \text{ KPa.} \\ E^*/E &= 0.486563 ; \nu^* = 0.297378 \end{aligned}$$

Skip Step 4 for Configuration d :

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$\begin{aligned} &= A / Do \\ &= 487.0/476.38 \\ &= 1.0223 \end{aligned}$$

Coefficient [F]:

$$\begin{aligned} &= (1 - \nu^*) / E^* \cdot (E \cdot \ln(K)) \\ &= (1 - 0.3) / 95607992 \cdot (0.19650E+09 \cdot \ln(1.02)) \\ &= 0.0318 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned} &= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt] \\ &= 476.38^2/16 \cdot [(1.066 - 1) \cdot (1.066^2 + 1) \cdot 25.511 - \\ &\quad (1.066 - 1) \cdot (1.066^2 + 1) \cdot 5.516] \\ &= 40230.3086 \text{ bars}\cdot\text{mm.}^2 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]

$$\begin{aligned} &= Mts + W^* \cdot (Gc - Gs) / (2 \cdot \pi \cdot Do) \\ &= 40230.3 + 73155.6 \cdot (508.0 - 508.0) / (2 \cdot \pi \cdot 476.38) \\ &= 40230.3086 \text{ bars}\cdot\text{mm.}^2 \end{aligned}$$

Note: W is the maximum of the bolt loads between the shell and channel sides.*

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$\begin{aligned} &= ((M^*) - Do^2/32 \cdot F(Ps - Pt)) / (1 + F) \\ &= ((40230.31) - 476.38^2/32 \cdot 0.032 (25.51 - 5.52)) / (1 + 0.03) \\ &= 34613.4648 \text{ bars}\cdot\text{mm.}^2 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned} &= Mp + Do^2/64 \cdot (3 + \nu^*) \cdot (Ps - Pt) \\ &= 34613.46 + 476.38^2/64 \cdot (3 + 0.297) \cdot (25.51 - 5.52) \\ &= 268398.7188 \text{ bars}\cdot\text{mm.}^2 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned} &= \max(| Mp | ; | Mo |) \\ &= \max(| 34613.465 | ; | 268398.719 |) \\ &= 268398.7188 \text{ bars}\cdot\text{mm.}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu^* (h - hg')^2) \\ &= 6 \cdot 268398.719 / (0.4442 (52.0 - 6.0)^2) \end{aligned}$$



$$= 171.3380 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot S \\ &= 2 \cdot 138.0 \\ &= 275.99 \text{ N./mm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 268123.0 / (0.4442 (42.2308 - 6.0)^2) \\ &= 275.9093 \text{ N./mm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [H_{reqB}]:

$$\begin{aligned} &= h + Cats + Catc \\ &= 42.2308 + 0 + 0 \\ &= 42.2308 \text{ mm.} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [H_{req}]:

$$\begin{aligned} &= \max(H_{reqB}; H_{reqS}) \\ &= \max(42.2308 ; 5.3929) \\ &= 42.2308 \text{ mm.} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$\begin{aligned} &= |P_s - P_t| \\ &= |25.51 - 5.52| \\ &= 19.995 \text{ bars} \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / Do \cdot \min(0.8 \cdot S; 0.533 \cdot S_y) \\ &= 4 \cdot 0.4 \cdot 52.0 / 476.38 \cdot \min(0.8 \cdot 137.997 ; 0.533 \cdot 222.731) \\ &= 19.281 \text{ N./mm}^2 \end{aligned}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$\begin{aligned} &= 1 / (4 \cdot \mu) \cdot (Do / \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot |P_s - P_t| + Cats + Catc \\ &= 1 / (4 \cdot 0.4) \cdot (476.38 / 110.398) \cdot |25.51 - 5.52| + 0 \\ &= 5.3929 \text{ mm.} \end{aligned}$$

Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [A_t]:

$$\begin{aligned} &= \pi (do - t) t \\ &= \pi (19.05 - 2.77) 2.77 \\ &= 1.417 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:

$$\begin{aligned} &= \pi \cdot t (do - t) S \\ &= \pi \cdot 2.77 (19.05 - 2.77) 92.398 \\ &= 1334.810 \text{ Kg} \end{aligned}$$

Groove Weld Strength [F_g]:

$$\begin{aligned} &= 0.85 \cdot \pi \cdot a_g (do + 0.67 \cdot a_g) S_w \text{ (but not } > F_t) \\ &= 0.85 \cdot \pi \cdot 2.77 (19.05 + 0.67 \cdot 2.77) 92.4 \\ &= 1334.8102 \text{ Kg} \end{aligned}$$

Weld Strength Factor [f_w]:

$$\begin{aligned} &= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\ &= 92.398 / 92.398 \\ &= 1.000 \end{aligned}$$

Design Strength Ratio [f_d]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [F_f]:



$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 2.77 (19.05 + 0.67 \cdot 2.77) 92.4$$

$$= 942.7503 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \cdot \sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd) - 0.75 \cdot do}$$

$$= 2 \cdot \sqrt{((0.75 \cdot 19.05)^2 + 1.07 \cdot 2.77 \cdot (19.05 - 2.77) \cdot 1.0 \cdot 1.0) - 0.75 \cdot 19.05}$$

$$= 3.198 \text{ mm.}$$

Minimum Required Fillet Weld Leg [afr]: 1.6620 mm.

Minimum Required Groove Weld Leg [agr]: 1.6620 mm.

Tube-Tubesheet Jt allowable, 1334.81 is >= tube strength 1334.81 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	171.3 <=	276.0 N./mm ²	Ok
Tubesheet shear stress	11.4 <=	110.4 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	42.231	52.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.662	2.770	Ok
Tube-Tubesheet Groove Weld Leg :	1.662	2.770	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend	Stresses Allwd	Shear	Allwd	Case Type	Pass/ Fail
D1uc	24.987	...	47	276	3	110	Ps+Pt D1	Ok
D2uc	47.023	28.013	219	276	15	110	Ps+Pt D2	Ok
D3uc	42.231	...	171	276	11	110	Ps+Pt D3	Ok
D1c	28.109	...	54	276	4	110	Ps+Pt-c D1	Ok
D2c	50.076	28.013	252	276	17	110	Ps+Pt-c D2	Ok
D3c	45.399	...	197	276	13	110	Ps+Pt-c D3	Ok
Max:	50.0765	28.013 mm.	0.912	0.151	(Str. Ratio)			

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.



Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	50.076	52.000	Ok
Tubesheet thickness flanged extension :	28.013	32.000	Ok
Tube thickness :	0.513	2.770	Ok
Tube-tubesheet fillet weld leg :	1.662	2.770	Ok
Tube-tubesheet groove weld leg :	1.662	2.770	Ok

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	27.970	1.000	27.970	1.000
Tubesheet Shear Stress	169.359	1.000	169.359	1.000
Tube Pressure Stress	304.060	1.000	222.141	1.000
Tubesheet Extension Stress	No Calc	No Calc	28.598	...
Minimum MAWP	27.970		27.970	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	32.207	1.000	32.207	1.000
Tubesheet Shear Stress	192.798	1.000	192.799	1.000
Tube Pressure Stress	304.060	1.000	222.141	1.000
Tubesheet Extension Stress	No Calc	No Calc	28.598	...
Minimum MAPnc	32.207		28.598	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Determine the governing MDMT considering the governing condition:

Governing thickness on the shell side per figure UCS-66.3 (c):

= tubesheet thickness/4

= 52.0/4

= 13.000 mm.

Thickness Ratio = 0.892, Temperature Reduction per Fig. UCS 66.1 = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -21 °C

Min Metal Temp. at Required thickness (UCS 66.1) -27 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:

= max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1

= max(5.52/32.21, 25.51/28.6)

= 0.892

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